Journal of Open Innovation: Technology, Market, and Complexity 11 (2025) 100521

Contents lists available at ScienceDirect

Journal of Open Innovation: Technology, Market, and Complexity

journal homepage: www.sciencedirect.com/journal/journal-of-open-innovation-technology-market-and-complexity

Not all forms of artificial intelligence are perceived equal: AI functions and work outcomes

Jieqiong Cao [a,*], Jingxian Yao [a], Shuhua Sun [b], Zhaoli Song [c], Fengzhi Zhang [d]

[a] School of Business, Singapore University of Social Science, Singapore
[b] A.B. Freeman School of Business, Tulane University, United States
[c] Department of Management and Organization, National University of Singapore, Singapore
[d] Board of Directors, Nanyang Global Innovation Institute, Singapore

ARTICLE INFO

Keywords:
Artificial intelligence
Perceived AI functions
Opportunity and threat appraisals
Cognitive appraisal theory

ABSTRACT

People hold mixed views about adopting artificial intelligence (AI) in the workplace – some believe that AI facilitates work processes, while others worry that it poses a threat to the importance of human labor. To address this conundrum regarding how employees feel and behave when AI is applied in their jobs, this study builds upon a typology of perceived AI functions and examines how these functions differentially impact employees' psychological appraisals and subsequently influence their attitudes and behaviors toward AI adoption. Specifically, AI functions are categorized into assistive AI, augmented AI, and autonomous AI. Drawing from cognitive appraisal theory and threat-rigidity theory, the research proposes that these functions are differentially related to opportunity and threat appraisals, which in turn affect employees' AI learning behavior and job insecurity. Furthermore, AI self-efficacy is hypothesized to function as a moderator that weakens the effects of augmented and autonomous AI on threat appraisal. Three studies using diverse samples and methodologies provide extensive empirical support for the hypotheses. Theoretical and practical implications of the findings are discussed.

Artificial intelligence (AI), generally defined as the ability of a machine or algorithm to learn from experience, adjust to new inputs, and perform human-like tasks (Duan, Edwards, and Dwivedi, 2019), has encroached on a wide array of industries and businesses. The number of organizations implementing AI has been increasing in the past five years (Chowdhury et al., 2022). For example, in healthcare, deep-learning-based AI models have been used to help pathologists improve the accuracy of breast cancer detection (Madani et al., 2022). In banking, companies have increasingly relied on AI to detect fraudulent transactions. In finance, AI has been widely used to automate trading, transactions, and portfolio management, which used to involve a lot of human effort. According to a JP Morgan survey, more than 60 % of trades in the equity market were performed by algorithms in the month of March 2020 (Lim, 2021). KPMG predicted that almost all companies will use AI for financial reporting in 3 years (Jalil, 2024). In manufacturing, Siemens has introduced SiePA, a predictive maintenance software that uses industrial big data analysis and machine learning to helps businesses anticipate and prevent equipment failures, improving operational efficiency (Siemens, 2024). In retail, companies like Amazon and Alibaba are using AI-driven search engines to enhance personalized recommendations, helping both sellers and customers find relevant products more efficiently (Knight, 2024). Meanwhile, logistics leaders like DHL are leveraging AI in supply chain management. By analyzing factors like weather conditions, traffic, and historical data, they can adjust transportation routes in real-time, reducing delays and costs (DHL Group, 2024).

However, the implementation of AI initiatives has not always yielded the anticipated gains and outcomes for companies (Ransbotham et al., 2019). According to a survey by the Boston Consulting Group and MIT, 70 % of AI projects failed to achieve their anticipated outcomes. One contributing factor of such failures is the difficulty employees face in integrating AI effectively into their existing workflows and processes (Makarius et al., 2020). Employees, as the direct users of and collaborators with AI, play a pivotal role in bridging AI as a technological tool with key organizational objectives (Lichtenthaler, 2019). The full potential of AI can only be realized when employees exhibit positive attitudes and responses towards AI adoption (Glikson and Woolley, 2020; Lichtenthaler, 2019). However, there is a notable absence of

* Correspondence to: School of Business, Singapore University of Social Sciences, 463 Clementi Road, 599494, Singapore.
E-mail address: jieqiongcao@suss.edu.sg (J. Cao).

https://doi.org/10.1016/j.joitmc.2025.100521
Received 13 December 2024; Received in revised form 13 March 2025; Accepted 19 March 2025
Available online 23 March 2025
2199-8531/© 2025 The Author(s). Published by Elsevier Ltd on behalf of Prof JinHyo Joseph Yun. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

theoretically grounded research models and empirical investigations on employees' reactions towards AI adoption (Chowdhury et al., 2022).

Employees form subjective perceptions of AI functions, and these perceptions shape their appraisals, attitudes, and behaviors toward AI. Different from previous automation technologies, AI has capabilities to think and act more like a human being (Čerka et al., 2015), which enables it to perform a broader set of tasks, ranging from repetitive tasks to human-like decision making (Manyika et al., 2017). Thus, it can be perceived to have multiple functions. In line with recent frameworks of AI-human partnership (Chowdhury et al., 2022; Makarius et al., 2020), this study examines the evolving roles of AI and human employees, emphasizing a shift in agency from humans to AI. To better categorize employees' perceptions of AI functions, a three-tier typology is introduced, distinguishing between *assistive, augmented, and autonomous AI*. This framework highlights how different AI functions may influence employees' attitudes and behaviors toward AI adoption.

This study integrates cognitive appraisal theory (CAT) (Lazarus and Folkman, 1984) and threat-rigidity theory (Barr and Glynn, 2004; Dutton and Jackson, 1987) to provide a comprehensive framework for understanding employees' responses to AI adoption. CAT explains how employees assess AI functions as either opportunities or threats. Specifically, assistive AI is expected to lead to opportunity appraisal, augmented AI to both opportunity and threat appraisals, and autonomous AI to threat appraisal. Threat-Rigidity Theory explains how these appraisals translate into behavioral reactions, such as AI learning behavior or job insecurity. The combination of two theories allows for a more holistic understanding of both the psychological and behavioral dimensions of AI adoption in the workplace. Additionally, employees' AI self-efficacy is considered a personal coping resource (Jerusalem and Schwarzer, 1992) that buffers the effects of autonomous AI and augmented AI on threat appraisal.

Our study intends to make several important contributions. First, our research contributes significantly to the burgeoning field of AI within management studies by offering a nuanced perspective on employees' attitudinal and behavioral responses to AI. This approach stands in contrast to the prevailing trend in current research, which often conceptualizes AI as a technological phenomenon. Second, this study develops and validates an AI typology consisting of assistive, augmented, and autonomous functions, demonstrating its utility in providing a more nuanced understanding of employees' varied reactions to AI adoption. This approach builds upon prior research that often relies on an oversimplified, one-dimensional conceptualization of AI. Additionally, the proposed typology aligns with recent theorization on AI-human partnership (Raisch and Krakowski, 2021), emphasizing that employees' perceptions of AI functions are closely intertwined with AI-human role divisions and significantly shape their experiences in technology-driven workplaces. Third, while several studies have identified factors influencing AI adoption (Morse, 2020; Tambe et al., 2019), researchers have raised concerns about the lack of a comprehensive theoretical framework for understanding AI-employee integration in organizational contexts (Chowdhury et al., 2022). To address this gap, this study integrates the principles of cognitive appraisal theory and threat-rigidity theory to examine the roles of opportunity and threat appraisals as mediators and AI self-efficacy as a moderator. This approach offers a theoretical explanation of how and when different AI functions influence key employee outcomes. Lastly, given that AI has been increasingly built into open innovation models and cultures (Broekhuizen et al., 2023), a deeper understanding of employees' perceived AI functions helps to assess and enhance workforce readiness for open innovation.

The following section provides a succinct analysis of current AI literature in the management domain, identifying its limitations in explaining employees' attitudes and behaviors toward AI. Based on these insights, a theoretical framework is proposed, categorizing AI functions into three types: assistive, augmented, and autonomous AI. The framework further hypothesizes the potential impact of these AI functions on employees' psychological appraisals, attitudes, and behaviors toward AI adoption. To test these hypotheses, three empirical studies were conducted: one to develop and validate a measure of perceived AI functions, an online experimental study to examine the causal effects of different AI functions on work outcomes, and an onsite two-wave field survey to test the full model.

1. Theory and hypotheses

1.1. AI functions

To study the impact of AI in organizations, it is essential to differentiate AI and AI functions. AI is an aggregation of several technologies, ranging from computer vision, natural language processing, to machine learning (Chui et al., 2018; Tredinnick, 2017). AI functions, instead, pertain to the roles AI plays in employees' work. AI receives training, learns from experiences, and thinks and acts like a human being (Čerka et al., 2015). With these skills, AI can perform a wider range of tasks, ranging from basic ones, such as repetitive physical tasks, to tasks that require human-level intelligence (Manyika, 2017), thus being perceived to have different functions.

Currently, there are several existing categories of AI functions. Baird and Maruping (2021) introduced four types of agentic information systems (IS) artifacts: reflexive, supervisory, anticipatory, and prescriptive, ranging from the lower end as assistant-like agents mainly assisting in routine tasks (i.e., reflective and supervisory artifacts), to the higher end of autonomous, capable of making complex decisions and operating independently (i.e., anticipatory and prescriptive artifacts) (Baird and Maruping, 2021). Another typology delineates AI applications into four categories: automated intelligence, which involves digital assistants automating routine tasks; assisted intelligence, where AI aids in human decision-making by summarizing data; augmented intelligence, which enhances human decision-making; and autonomous intelligence, where AI systems operate independently without human intervention (Chowdhury et al., 2022; Daugherty et al., 2019; Davenport and Bean, 2018; Haenlein and Kaplan, 2019). Raisch and Krakowski (2021) also divided AI systems into two primary functions: AI automation, where AI takes over tasks typically performed by humans, and AI augmentation, where AI systems or machines surpass human cognitive capabilities.

The above categorizations share similarities in that they are built on the agency of both employees and AI within tasks. Building on these existing frameworks, this study introduces a typology of AI functions with three distinct categories: assistive, augmented, and autonomous. This typology represents a spectrum of AI roles, ranging from assisting employees in tasks to enhancing their capabilities and ultimately performing tasks independently. Functionally, for assistive AI, employees retain the final say, with AI executing tasks according to their directives. It also acts as a substitute for repetitive human labor. For augmented AI, the role of AI expands from mere execution to enhancement, enabling the performance of tasks beyond human capabilities, such as extracting insights and making predictions from extensive datasets. In contrast, for autonomous AI, the role of employees shifts from being decision-makers to implementers, executing tasks based on AI-generated outputs.

Overall, our typology adds to existing categorizations by cautioning against oversimplifying AI as a unified phenomenon and highlighting the subjectivity in perceiving the roles of AI. For instance, the role of ChatGPT, the large language AI implication developed by OpenAI, may be perceived differently across individuals. Some individuals view ChatGPT as assistive, focusing on its role in simplifying tasks like data retrieval or generating templated responses. Others perceive it as an autonomous entity capable of creating unique contents or performing evaluations based on input parameters without human oversight. Additionally, there are those who consider ChatGPT as augmented AI. This is evident in tasks like complex data analysis, where ChatGPT not only processes large data sets but also uncovers patterns and trends that might elude human analysts. Taken together, our typology of AI

functions speaks effectively to how employees view AI adoption in their jobs.

2. Cognitive appraisal theory (CAT)

CAT suggests that individuals, when confronted with external events or situations, engage in a two-stage appraisal process that includes primary appraisal and secondary appraisal (Lazarus and Folkman, 1984). In primary appraisal, they assess whether an event or situation is relevant to themselves, and if so, whether it is harmful or beneficial (Lazarus, 1991; Lazarus and Folkman, 1987). If the event is deemed relevant, the secondary appraisal process determines whether individuals have the ability and resources to overcome them. The appraisal result, in turn, affects people's subsequent attitudes and behaviors (Lazarus and Folkman, 1984). Researcher later extended CAT to include additional appraisal dimensions such as novelty, goal congruence, and agency, these dimensions further refine how individuals perceive and react to external stimuli. Specifically, novelty captures the extent to which an event is perceived as new or unexpected, goal congruence reflects whether an event aligns with an individual's personal or professional objectives, while agency determines whether individuals attribute control over an event to themselves or an external force (Roseman et al., 1990; Smith and Ellsworth, 1985). These new dimensions makes CAT particularly relevant in AI adoption, as they may closely influence how employees interpret AI functions and form subsequent responses.

Scholars have used the cognitive appraisal lens to explain a variety of phenomena, such as work-related stressors (Fisher et al., 2013; Norman et al., 1995) and undesirable family situations (Lim and Tai, 2014). More relevant to the current research context, prior research has applied CAT to examine how employees interpret and respond to workplace technologies, particularly in contexts of digital transformation. Beaudry and Pinsonneault (2005) developed a framework integrating CAT to identify adaptation strategies that influence individual responses to IT adoption, highlighting the role of control in influencing whether technology is appraised as a challenge or a threat. Similarly, Johnston and Warkentin (2010) applied CAT to cybersecurity contexts, demonstrating that employees' perceptions of technological risks and benefits influence their behavioral responses. Liang and Xue (2009) and Liang et al. (2019) explored how individuals' cognitive appraisals of technology adoption shape their responses, particularly in security and information system contexts. These findings suggest that technology-driven changes in the workplace can induce both opportunity and threat appraisals, depending on perceived control and familiarity. These studies provide foundation for applying CAT to AI adoption, as AI functions (assistive, augmented, and autonomous) introduce varying levels of control and uncertainty that influence employees' appraisals. By integrating insights from prior CAT-based research, this study advances our understanding of how employees psychologically and behaviorally respond to AI in workplace settings.

Another key reason for selecting CAT is that it explains how individuals interpret and respond to external changes in their work environment. AI adoption presents a dual reality—both an opportunity and a threat—which makes CAT particularly suitable for capturing employees' cognitive and emotional responses. While alternative theories, such as Technology Acceptance Model (TAM), offer insights into employee acceptance and usage of technologies, they do not fully capture how employees cognitively and emotionally appraise AI adoption. In contrast, CAT is uniquely positioned to explain the thought processes and emotions that shape employees' reactions to AI in the workplace.

Upon facing AI adoption in the workplace, employees may evaluate whether the adopted AI is favorable or unfavorable based on the perceived roles of AI in their work. In this research, we focus on opportunity appraisal and threat appraisal. Opportunity appraisal refers to individuals' positive judgment that they can benefit from external events or situations and obtain gains (Dutton and Jackson, 1987; Zhou et al., 2008). It represents employees' overall evaluation that the adopted AI facilitates their work processes or improves their work experiences. Threat appraisal refers to the judgment that the events or situations have the potential to inhibit their thriving and generate harmful consequences (Lazarus and Folkman, 1987), and in this study, it represents employees' global negative evaluation of adopted AI for their work.

Employees are likely to use several evaluation criteria to form opportunity and threat appraisals. For example, when employees feel productive and motivated (Sterling and Boxall, 2013), experience autonomy (Morgeson et al., 2005), engage in enriching tasks (Parker, 1998), or have learning and development opportunities (Shuck et al., 2013), they are likely to develop opportunity appraisal. In contrast, when employees feel incompetent (Von Hippel et al., 2005), lose a sense of control and agency (Bond and Bunce, 2003), or experience a crisis in self-identity or self-value (Aquino and Douglas, 2003), they tend to develop threat appraisal. Combining the criteria of opportunity and threat appraisal with the categorization of three AI functions, we expect that assistive AI will lead to opportunity appraisal, autonomous AI will lead to threat appraisal, and augmented AI will lead to opportunity and threat appraisal simultaneously.

3. Assistive AI and opportunity appraisal

In assistive AI, employees retain ultimate control, with AI executing the tasks directed by employees. This form of AI replaces repetitive human labor. Assistive AI may be appraised as an opportunity for the following reasons. To begin with, from the perspective of CAT's goal congruency dimension, assistive AI aligns with employees' objectives by enhancing task completion and improving efficiency. With AI executing the tasks that used to be done by employees, AI could facilitate employees' work progress and free them to craft their tasks, which can benefit employees (Zhang and Parker, 2019) or engage in tasks that can energize them (Wilson and Daugherty, 2018). In other words, employees can spend less time and resources, especially in repetitive labor, and achieve equivalent or better work outputs. For example, in a survey of doctors, Sinsky et al. (2016) found that AI can help with clerical tasks, reduce the burden of administrative jobs by scheduling operating rooms, ordering tests, and prescribing medications, and thus let doctors concentrate more on their interactions with patients. Further, assistive AI may provide avenues for employees' growth and development as it can be viewed as a chance to restructure their capabilities (Jaiswal et al., 2022; Wilson and Daugherty, 2018). Taken together, assistive AI is likely to make employees form overall positive evaluations of AI as it can help enhance their work efficiency and focus on important tasks, job meaningfulness, and personal development. We propose:

Hypothesis 1. : Assistive AI will be positively related to opportunity appraisal.

3.1. Autonomous AI and threat appraisal

With autonomous AI, decision-making power shifts from employees to AI, while employees primarily execute tasks based on AI-generated instructions or suggestions. This transfer of control may activate employees' threat appraisal. First, from a CAT perspective, autonomous AI introduces a novel work environment in which employees experience a loss of agency and control, thereby triggering threat appraisal. Skavronskaya et al. (2020) suggest that novelty heightens attention and memory toward the target of appraisal, meaning employees are more likely to focus intensely on novel AI functions. However, unlike in Skavronskaya et al.'s (2020) study, where novel experiences tend to be reconstructed positively over time, the novelty of autonomous AI may instead amplify its perceived agency-threatening nature. When employees first encounter autonomous AI, they may fixate on its capacity to replace human decision-making, reinforcing threat appraisal (Bond and

Bunce, 2003). Research further supports these concerns. Studies show that losing control is a key factor underlying people's reluctance to rely on algorithms (Dietvorst et al., 2016) and that work environments perceived as limiting employees' control tend to be appraised as harmful and threatening (Bosma et al., 1998; Vander Elst et al., 2014).

Second, employees place importance on their identity and value their importance at work (Hackman and Oldham, 1975). If the adopted AI can accomplish work tasks with cognitive capabilities beyond (or believed to be beyond) what employees possess, employees may perceive a less unique identity and lower value at work (Castelo, 2019), which can spontaneously elicit threat appraisal (Aquino and Douglas, 2003). In support, consumer behavior research has shown that people resist automated products with amplified functions, because these functions hinder their positive feelings and unique identities as product users, especially in domains they deem important (Leung et al., 2018). Moreover, research has shown that when the adopted AI has substitutive decision-making features, employees try to protect their professional role identity even when they do not use the outputs or decisions from the AI system (Strich et al., 2021). We propose:

Hypothesis 2. : Autonomous AI will be positively related to threat appraisal.

3.2. Augmented AI and opportunity and threat appraisals

In augmented AI, the role of AI expands from execution to extension by performing tasks that humans cannot do, yet employees' decision-making power is reserved. We argue that augmented AI can trigger employees' opportunity and threat appraisal simultaneously.

Augmented AI opens new and better ways of doing things and enables employees to perform a broader range of tasks, including challenging tasks they are unable to do without the help of augmented AI. Employees innately favor challenging and meaningful tasks, as they believe these tasks can benefit them in the long term (VandeWalle et al., 1999). An example is when augmented AI can capture the complexity of data, build relationships among variables, and present an overview of the information streams (Jarrahi, 2018), thus providing employees with solid (and otherwise unattainable) bases for making decisions. This echoes with the dimension of goal congruence of CAT in that augmented AI facilitate employees' goal of task completion. Therefore, employees may consider the adoption of augmented AI as an opportunity to enhance their work effectiveness.

However, at the same time, the adoption of augmented AI may cause employees' negative evaluations of AI. As AI penetrates more aspects of work that exceed employees' abilities, they may gradually develop the belief that they are inferior to AI. In addition, AI is oftentimes lacking in the capacity to explain how it arrives at specific predictions, which emerges as a black box and keeps employees in the dark (Chui et al., 2018). The opacity of augmented AI may devalue and discourage employees, as they oftentimes find it hard to make sense of AI outputs. As evidenced by Zuboff (1988), the opacity of the information system in banking procedures makes employees less informed about their financial knowledge and work. Moreover, with the gradual uncertainty caused by the technologies and competitiveness within organizations, employees are vigilant about their status in organizations, and actively search for information that may threaten their status (Reh et al., 2018). In the context of AI adoption, augmented AI introduces a novel and uncertain workplace dynamic, potentially leading employees to feel a loss of agency and control over their work, thereby eliciting negative reactions toward the adopted AI. Therefore, employees may simultaneously develop opportunity and threat appraisals with augmented AI. We propose:

Hypothesis 3. : Augmented AI will be positively related to both (a) opportunity and (b) threat appraisal.

3.3. Implications on work outcomes

Cognitive appraisal theory posits that people will react to situations or events after they appraise them. Once employees evaluate AI functions as an opportunity or threat, they engage in secondary appraisal, in which they assess their personal resources and abilities to cope with the AI-driven changes (Lazarus and Folkman, 1984). Employees who perceive themselves as capable of adapting to AI may engage in problem-focused coping. Conversely, employees who lack AI-related confidence may adopt emotion-focused coping. Integrating the cognitive appraisal process with threat-rigidity theory, we explore employees' reactions in response to opportunity and/or threat appraisals of AI functions. Threat-rigidity theory suggests that threatening situations restrict individuals' information processing and narrow their attention (Staw et al., 1981). It has been used in traditional technology contexts to explain why employees or organizations perform poorly when facing rapid technology changes (Joseph and Ang, 2001; Ketchen Jr and Palmer, 1999). Following the basic tenets of the threat-rigidity perspective, researchers further developed this theory by incorporating both negative and positive situations and discussed how positive situations broaden attention and thought-action repertoires (Barr and Glynn, 2004; Dutton and Jackson, 1987). Opportunity and threat appraisals capture employees' perceptions of the extent to which AI provides benefits or imposes threats on them, respectively (Kraimer et al., 2005). The perceptions of favorable versus unfavorable situations brought by AI, we argue, explain employees' subsequent reactions toward the adoption of AI. Specifically, we focus on employees' adaptive behaviors for AI adoption (i.e., AI learning behavior), and their sense of powerlessness to maintain their job status (i.e., job insecurity).

We propose that opportunity appraisal facilitates AI learning behavior after AI adoption. Based on CAT, employees who perceive AI as the opportunity may engage in problem-focused coping, such as actively learning AI-related skills or seeking training (Beaudry and Pinsonneault, 2005). Based on extensions to the threat rigidity theory (Barr and Glynn, 2004; Dutton and Jackson, 1987), if an external situation is appraised to be potentially positive, approach-oriented behaviors are likely to occur, as manifested by allocating more efforts to learning. Indeed, researchers have argued that learning, as a process of accumulating knowledge and correcting errors (Argyris and Schön, 1997), is more likely to take place when individuals foresee the positive relationship between their efforts and the actual learning process (Vroom, 1964). Opportunity appraisal, which promotes employees' feelings of mastery and confidence (Baldwin and Magjuka, 2014; Mathieu and Martineau, 2014; Maurer, 2001), is likely to increase learning behavior.

Opportunity appraisal, at the same time, is likely to lower employees' job insecurity as employees capable of adapting to AI are less likely to adopt emotion-focused coping strategies (Folkman and Lazarus, 1985). Job insecurity is defined as perceived powerlessness to maintain desired continuity in one's job (Greenhalgh and Rosenblatt, 1984) and "overall concern about the future existence of the job" (Rosenblatt and Ruvio, 1996, p. 587). It captures employees' perceived inability to maintain their current job status. As mentioned above, assistive and augmented AI provide employees with the perception that they can benefit from AI adoption, either because they can work more efficiently and focus on more meaningful tasks or because AI extends their capabilities to perform their jobs more effectively. Opportunity appraisal, based on such positive construal of AI, provides employees with the feeling that they can create more value for the organization, and, because of that, they have better prospects in their jobs. We hypothesize:

Hypothesis 4. : Assistive AI will have a positive and indirect effect on (a) AI learning behavior, and a negative and indirect effect on (b) job insecurity via opportunity appraisal.

Hypothesis 5. : Augmented AI will have a positive and indirect effect on (a) AI learning behavior, and a negative and indirect effect on (b) job insecurity via opportunity appraisal.

Threat appraisal refers to employees' negative evaluations of the adopted AI and occurs when employees perceive external situations as harmful or unfavorable (Britton et al., 2011). Threat appraisal may restrict AI learning behavior for two reasons. First, according to threat-rigidity theory (Staw et al., 1981), if employees perceive a situation as threatening, they will narrow their attention to the dominant cues of the situation, rather than attend to diverse information and approach novel knowledge to improve their capabilities. Even when employees attempt to engage in AI learning behavior, the negative perceptions and biases against AI may restrict their information processing and inhibit their learning (Staw et al., 1981). Second, when threat appraisal is stimulated, employees tend to engage in emotion-focused coping (Staw et al., 1981; Zhou et al., 2008), and their problem-focused coping, such as active learning, will be demotivated or restricted (Folkman and Lazarus, 1985). As such, employees are less likely to broaden their minds to learn new information both about their reformed tasks and about the introduced AI.

Threat appraisal can also increase job insecurity. Job insecurity happens when there is the involuntary loss of a job or important job features (Ashford et al., 1989; Greenhalgh and Rosenblatt, 1984). Employees usually transform organizational information into a perceptual process and analyze whether the situation will cause potential loss. Factors that can result in loss of control and attendant anxiety are the sources of job insecurity (Ashford et al., 1989). We argue that threat appraisal will engender job insecurity for two main reasons. First, a key component of job insecurity is the severity of loss one experiences in the job (Greenhalgh and Rosenblatt, 1984), and threat appraisal, as a global negative evaluation of AI, reflects that desirable aspects of the work environment have been taken away. Put differently, threat appraisal about AI can be extended to a threat to one's job and its desired features, thereby triggering a sense of job insecurity. Second, threat appraisal in our research context captures a loss of control and self-value at work, both of which are typical sources of job insecurity (D'Souza et al., 2003; Lee and Peccei, 2007). We thus hypothesize:

Hypothesis 6. : Augmented AI will have a negative and indirect effect on (a) AI learning behavior, and a positive and indirect effect on (b) job insecurity via threat appraisal.

Hypothesis 7. : Autonomous AI will have a negative and indirect effect on (a) AI learning behavior, and a positive and indirect effect on (b) job insecurity via threat appraisal.

4. AI Self-efficacy as moderator

According to CAT, secondary appraisal plays a critical role in shaping employees' reactions to AI. This process determines whether individuals perceive themselves as having the ability and resources to develop effective coping strategies and navigate potentially threatening situations. When individuals appraise stressful external events, they evaluate their personal resources and capabilities to manage the situation (Lazarus and Folkman, 1987). In this regard, personal resources function as a buffer, mitigating the adverse effects of environmental stressors on threat appraisals (Lazarus and Folkman, 1987). Among these personal resources, self-efficacy has been widely recognized as a critical psychological buffer (Bandura, 1997; Jerusalem and Schwarzer, 1992; Heuven et al., 2006). Prior research suggests that self-efficacy can help individuals regulate negative emotional reactions to technological changes (Johnston and Warkentin, 2010). Extending this perspective to AI adoption in the workplace, we focus on AI self-efficacy as a key determinant in mitigating the potential negative effects of AI functions on employees' psychological and behavioral responses.

Self-efficacy is generally defined as the belief that a person has the capabilities needed to succeed at a specific task (Bandura, 1997). According to Bandura (1997), self-efficacy is a domain-specific belief and carries a clearer meaning when contextualized. In this research, we extend the original concept of self-efficacy to the context of AI and conceptualize a domain-specific construct, AI self-efficacy, which refers to one's belief in being able to use AI to accomplish tasks and fulfill role expectations. Invoking the resource-buffering hypothesis (Jex et al., 2001; Jerusalem and Schwarzer, 1992), we expect that AI self-efficacy weakens the effects of augmented AI and autonomous AI on threat appraisal.[1]

We posit that AI self-efficacy buffers the effect of augmented AI on threat appraisal and downstream consequences of AI learning behavior and job insecurity for several reasons. As discussed, augmented AI threatens employees due to its ability to perform tasks beyond human cognition. Employees with high AI self-efficacy hold strong beliefs about their capabilities in coping with unpredictable situations brought by augmented AI and facilitating integration between AI and human intelligence. Furthermore, employees with high AI self-efficacy tend to focus more on the benefits of augmented AI, rather than the challenges that it brings (Makara-Studzińska et al., 2019), thus to a lesser extent viewing augmented AI as a threatening situation. We propose:

Hypothesis 8. : AI self-efficacy will moderate the positive relationship between augmented AI and threat appraisal, such that the relationship will be weaker when AI self-efficacy is high.

Hypothesis 9. : AI self-efficacy will moderate the indirect relationships between augmented AI and (a) AI learning behavior and (d) job insecurity via threat appraisal, such that the indirect relationships will be weaker when AI self-efficacy is high.

Autonomous AI makes decisions independently without human intervention, which may lead to employees' fear of losing control and being replaced by AI. We expect that AI self-efficacy will buffer the effect of autonomous AI on threat appraisal and downstream outcomes for the following reasons. First, AI self-efficacy may change how employees evaluate their work situation where AI is adopted (Chemers et al., 2000; Hsieh et al., 2007). With high AI self-efficacy, employees may consider the application of AI as a trigger that propels them to grasp new knowledge or expertise, instead of replacing them and downgrading their status. This is likely to make them less distracted by or anxious about the adopted AI. Second, AI self-efficacy may motivate employees to take more adaptive responses to the undesired situation (Kinicki and Latack, 1990; Jex and Bliese, 1999). Employees with high AI self-efficacy are problem-solving-oriented. They may try to seek and integrate extra information and re-clarify or craft their job responsibilities to get a reconciled state (Brown et al., 2001) instead of being immersed in a threatened state. We propose:

Hypothesis 10. AI self-efficacy will moderate the positive relationship between autonomous AI and threat appraisal, such that the relationship will be weaker when AI self-efficacy is high.

Hypothesis 11. AI self-efficacy will moderate the indirect relationships between autonomous AI and (a) AI learning behavior and (b) job insecurity via threat appraisal, such that the indirect relationships will be weaker when AI self-efficacy is high.

5. Overview of studies

To thoroughly investigate the impact of AI functions on employees' psychological appraisals and work outcomes, this research adopts a

[1] We do not predict the moderating role of AI self-efficacy in the assistive AI/augmented AI-opportunity appraisal relationship for two reasons. First, according to cognitive appraisal theory, when a favorable situation happens, individuals may not go into the second stage of appraisal and may not feel the need to assess their individual resources. Second, both assistive AI and augmented AI may lead to opportunity appraisal regardless of self-efficacy level. This is because, even for employees with low AI self-efficacy, if AI can enhance their work efficiency and free them for more energizing work tasks, they are expected to consider the adopted AI as instrumental and helpful.

multi-study design. The decision to conduct three distinct studies rather than a single comprehensive study is grounded in both theoretical and methodological considerations. Theoretically, a progressive validation process was essential for refining the conceptualization and measurement of AI functions before testing their effects on workplace outcomes. Methodologically, separating the studies allows for a more rigorous examination of causal relationships and external validity. Study 1 was conducted to establish and validate the typology of assistive, augmented, and autonomous AI functions. Study 2 employs an experimental design to establish causality between AI functions, cognitive appraisals, and work outcomes, while Study 3, a two-wave field study, extends the findings to employees working with AI in real-world settings, enhancing external validity and generalizability. The three studies can complement one another in terms of methodological (dis)advantages, while one single study can hardly cover all desired elements (e.g., causality and real-world applicability). Although a multi-study approach requires a longer research timeline, this design enhances the validity and applicability of the findings. The sequential structure enables a nuanced investigation of AI adoption in the workplace, allowing for both theoretical refinement and empirical validation across different methodological contexts.

6. Study 1 development of AI function scale

6.1. Study 1a: item generation

Given the lack of validated measures representing different AI functions, a systematic scale development process was conducted following the procedure recommended by Hinkin (1995). Items were deductively generated by first establishing a clear conceptual definition of the three AI functions. A total of 12 items were developed for assistive AI, 14 for augmented AI, and 14 for autonomous AI. To assess content validity, a panel of subject matter experts consisting of 19 Ph.D. students specializing in Organizational Behavior was invited. These experts were provided with definitions of assistive AI, augmented AI, and autonomous AI, along with the generated items, and were asked to evaluate the extent to which the items reflected the definitions of the three AI functions using a scale from 1 (strongly disagree) to 7 (strongly agree). Following the threshold of the Content Validity Index (CVI), items with a CVI below.78 were eliminated (Almanasreh et al., 2019). After this step, the remaining numbers of items for assistive AI, augmented AI, and autonomous AI were 7, 8, and 12, respectively.

6.2. Study 1b: exploratory factor analysis (EFA)

EFA is typically required in the initial stages of item development to eliminate poorly performing items (Hinkin, 1995). Using the set of 27 items, we recruited 200 participants through Prolific, a widely used online research platform that enables access to diverse and high-quality respondents. Given the study's focus on AI adoption in the workplace, several screening criteria were applied to ensure participant relevance and data quality. First, participants were asked whether they engaged with AI-related technologies in their work, and those who responded "No" were excluded. To further verify their AI involvement, they were required to describe the AI technologies they use in their work. These responses were manually reviewed, and any irrelevant or incomplete answers were screened out. Additionally, participants' Prolific IDs were monitored to prevent duplicate submissions, ensuring that no individual could participate more than once. To maintain data integrity, attention-check questions were embedded throughout the survey, and participants who failed these checks were removed from the final dataset. The final sample consisted of professionals from various industries who actively work with AI technologies, enhancing the study's external validity. After these measures, our final sample size comprised 176 participants. The demographics of our participants revealed an average age of 37.56 years (SD = 9.36), with 51.7 % identifying as female. Among the participants, 63.6 % hold a bachelor's degree or higher. 73.5 % identified themselves as being at the management level, while only 26.5 % reported being at the entry or junior level. The majority of participants work in sectors such as professional, scientific, or technical services, educational services, healthcare or social assistance, and finance or insurance. Furthermore, 71.5 % indicated that AI has been integrated into their work for more than six months.

Participants indicated the extent to which the 27 items aligned with the AI implemented in their work, using a scale from 1 (strongly disagree) to 7 (strongly agree). After collecting the data, we conducted a principal components EFA and set the number of factors to three with Promax rotation. The 27 items were then categorized into these three anticipated dimensions. We identified the items with the highest loadings in each of the three factors and removed those with cross-loadings. Following this refinement, the number of remaining items for assistive AI, augmented AI, and autonomous AI stood at 6, 5, and 8, respectively (see Table 1 for these items and their factor loadings). The results suggest that our AI function measure has both content adequacy and appropriate factor loadings and comprises three empirically distinct subfactors.

6.3. Study 1c: confirmatory factor analyses (CFA)

We subsequently conducted a CFA to validate the factor structure of our three-dimensional AI function measure. 200 participants were recruited following the same procedure as in the EFA study to ensure that all participants actively engaged with AI technologies in their work. The average age of the participants was 35.94 years (SD = 9.86). Gender distribution was balanced, with 49.5 % identifying as female. In terms of educational background, 75 % held a bachelor's degree or higher. Regarding job roles, 72.4 % identified as being in management

Table 1
EFA Results (Study 1b).

Item	Factor Loading
Assistive AI	
The adopted AI is within the scope of my capabilities.	.85
The tasks performed by AI are within my control.	.80
I am able to control what the adopted AI is doing.	.75
The adopted AI operates within the boundaries of my skill set.	.79
The adopted AI operates within the range of my abilities and expertise.	.71
The adopted AI performs tasks within my capabilities.	.70
Augmented AI	
The adopted AI explores alternative options that I wouldn't have considered otherwise.	.79
The way the adopted AI performs work tasks is beyond what I can do.	.69
The adopted AI provides new approaches to work tasks with its advanced capabilities.	.75
The adopted AI offers me recommendations based on its findings which I am unable to generate by myself.	.77
The adopted AI expands my thoughts beyond my current knowledge base.	.84
Autonomous AI	
The adopted AI can navigate intricate scenarios with minimal or no input from me.	.79
The adopted AI can make decisions without my intervention.	.73
The adopted AI can take full control of specific tasks without human intervention	.77
The adopted AI can make decisions in real-time, navigating complex situations with minimal or no human input.	.81
The adopted AI exercises decision-making capabilities independently of my input.	.80
The adopted AI carries out tasks without requiring continuous guidance from me.	.73
The adopted AI can make its own decisions in complex environments.	.75
The adopted AI is capable of executing actions in the absence of my oversight.	.77

Note: N = 176

positions, while only 27.6 % reported working at the entry or junior level. The participants primarily worked in industries such as professional, scientific, or technical services, healthcare or social assistance, finance or insurance, and information technology. Additionally, 76.5 % of respondents indicated that AI had been integrated into their work for more than six months, reflecting their sustained exposure to AI-driven workplace environments.

Participants completed an online survey containing 19 items. To minimize the impact of the sequence, the order of these items was randomized for each participant. We assigned the 19 items into three dimensions, and the three-factor model resulted in a satisfactory fit, with χ^2 (149) = 363.77, (p < .001), CFI = .90, RMSEA = .09, SRMR = .07 (Hu and Bentler, 1999). For each factor, the items with the top five highest loadings were retained. Following this step, the model fit further improved, with χ^2 (87) = 182.98, (p < .001), CFI = .94, RMSEA = .07, SRMR = .06. Importantly, the three-factor model with 15 items exhibited a better model fit compared to alternative models (see Table 2 for a summary). For instance, when assigning all the items to one factor, the model fit was poor, with χ^2 (90) = 1106.05, (p < .001), CFI = .37, RMSEA = .24, SRMR = .23.

6.4. Study 1d: validity testing

The discriminant validity of the AI function scale was examined by incorporating existing measures related to AI, such as AI familiarity, AI attachment, and trust stance in AI (Chi et al., 2021). Additionally, measures of threat appraisal and opportunity appraisal were included to test the criterion-based validity of the AI functions measure.

A total of 200 participants were recruited via Prolific, following the same screening procedure used in prior studies to ensure their involvement with AI technologies in the workplace. The participants had an average age of 37.50 years (SD = 10.99), with 49.5 % identifying as female. In terms of educational attainment, 73.7 % held a bachelor's degree or higher. Regarding job positions, 72.2 % were in management roles, while 27.8 % were at the entry or junior level. The majority of the participates are from industries including professional, scientific, and technical services, educational services, healthcare or social assistance, and finance or insurance. 70.2 % reported that AI had been integrated into their work for more than six months.

The participants completed an online survey comprising the 15-item AI function scale, a 4-item measure of AI familiarity, a 3-item measure of AI attachment, a 3-item trust stance in AI scale (Chi et al., 2021), a 6-item opportunity appraisal scale (Kraimer et al., 2005), and a 3-item threat appraisal scale (McGregor and Elliot, 2002). All these measures are validated and have demonstrated adequate reliabilities, ranging from.82 to.96.

We assessed whether our AI function measure or its dimensions loaded onto a single factor with other AI-related measures, including AI familiarity, AI attachment, and trust stance in AI, or onto separate factors. Comparing the fit of the two models for each AI function dimension, we found that the separate factors model provided a better fit with the data than the single factor model did ($\Delta\chi2$ = 2118.54, p < .001). This result provides evidence for the discriminant validity of the AI function as a whole and for each of its dimensions.

We then conducted regression analyses to assess whether our AI function measure explained variance in individuals' opportunity and threat appraisals, as predicted by our theory. Table 3 presents the descriptive statistics and correlations, and Table 4 shows the regression results. The regression results demonstrated that the three AI functions contributed significantly to explaining this variance. Specifically, assistive AI significantly explained variance in individuals' opportunity appraisal (b =.21, SE =.07, p < .01) and threat appraisal (b = −.30, SE =.11, p < .01); augmented AI also significantly explained variance in opportunity appraisal (b =.40, SE =.06, p < .01); and autonomous AI explained variance in threat appraisal (b =.20, SE =.09, p < .01). Through the validation efforts, the 15 items constituted our assistive-augmented-autonomous AI functions measure (5 for each AI function). These items, together with the items for other focal variables of this research, are presented in Appendix A.

Study 1 indicates that our AI functions measure encompasses three subdimensions – assistive, augmented, and autonomous AI. The discriminant validity test demonstrates that our scale possesses distinct values separate from existing constructs related to AI. Furthermore, the predictive criterion validity test shows that our operationalization of the AI function predicts individuals' opportunity and threat appraisals effectively, except for the nonsignificant relationship between augmented AI and threat appraisal.

Study 1 validated the scales of the three AI functions and demonstrated initial evidence of the relationships between AI functions and opportunity and threat appraisals. Since the data on AI functions and appraisals were collected simultaneously, there may be concerns regarding reverse causality and common method variance. To address these issues, a scenario-based experimental approach was employed in Study 2 to enhance causal interpretations. Additionally, distal outcomes were collected to examine the indirect effects of AI functions on these outcomes through appraisals.

Table 2
CFA Results (Study 1c).

Model	χ^2	df	CFI	RMSEA	SMER
Three-Typology AI function	182.98	87	.94	.07	.06
Two-Typology AI function[a]	595.19	89	.78	.16	.18
Two-Typology AI function[b]	1144.68	89	.55	.23	.23
Two-Typology AI function[c]	656.81	94	.76	.16	.21
One-Typology AI function[d]	1106.05	90	.38	.27	.26

Note. N = 200.
TLI = Tucker-Lewis index; CFI = confirmatory factor index
SRMR = standardized root mean square residual.
[a] This model combines assistive AI and augmented AI into one factor.
[b] This model combines assistive AI and autonomous AI into one factor.
[c] This model combines augmented AI and autonomous AI into one factor.
[d] This model combines all measurement items into one factor.

7. Study 2 method

7.1. Participants and design

Participants for this study were full-time employees recruited through Prolific. Initially, 233 individuals participated in the experiment. Considering that this study employed an experimental design in which participants were randomly assigned to one of the three AI function conditions, prior work experience with AI was not a prerequisite for participation. As a result, no screening questions were used during recruitment. Instead, attention check questions were placed at the end of the experiment to ensure data quality. Participants who failed these checks were excluded, resulting in a final sample of 218 individuals. Among the final sample, 55.1 % of participants were male, and the average age was 39.56 years (SD = 10.44). Participants were randomly assigned to one of three AI conditions, with 75 in the assistive AI group, 72 in the augmented AI group, and 71 in the autonomous AI group. Among them, only 3.7 % reported being unfamiliar with new technologies and innovations, suggesting that most participants had at least some level of technological competence.

7.2. Procedure and manipulation

Participants were invited to assume the role of an HR associate in a multinational corporation. This role primarily focuses on recruitment, alongside responsibilities in newcomer training and performance management, with recruitment being the pivotal aspect. The scenario presented to participants described a situation where the HR associate (a role they were enacting) was tasked by their HR manager with recruiting

Table 3
Descriptive Statistics and Correlations (Study 1).

	Mean	*SD*	1	2	3	4	5	6	7	8
1.Gender	1.49	.50	-							
2.Age	37.51	10.99	−.03	-						
3.Education	4.17	1.29	.08	.01	-					
4.Assistive AI	5.33	1.10	−.02	.05	.15 *	.90				
5.Augmented AI	4.71	1.29	.01	−.07	−.08	.06	.87			
6.Autonomous AI	4.33	1.45	−.01	−.12	.01	.03	.37 * *	.91		
7.Opportunity appraisal	5.15	1.22	−.05	−.05	.03	.22 * *	.45 * *	.23 * *	.91	
8.Threat appraisal	3.02	1.68	.04	−.07	.07	−.18 * *	.00	.16 *	−.47 * *	.97

Note. $N = 200$. *SD* = standard deviation.
Gender: 1 = male, 2 = female.
Reliability coefficients appear along the diagonal.
$^*p < .05$. $^{**}p < .01$.

Table 4
Summary of Regression Results (Study 1).

	Opportunity Appraisal		Threat Appraisal	
	b	*SE*	*b*	*SE*
Gender	−.13	.15	.08	.23
Age	.00	.01	−.01	.01
Education	.04	.06	.12	.09
Assistive AI	.21 * *	.07	−.30 * *	.11
Augmented AI	.40 * *	.06	−.07	.10
Autonomous AI	.05	.06	.20 *	.09
R^2	.25 * *		.08 * *	

Note. $N = 200$. *SE* = standard error.
Unstandardized regression coefficients are reported. $^*p < .05$. $^{**}p < .01$.

employees for five different positions, each having 2–10 openings.

After the job postings were made online, the HR associate received a diverse range of applications, numbering between 30 and 100 per position. These applications included resumes and additional supporting materials. The participants, stepping into the shoes of the HR associate, were then presented with a sample resume from one of the applicants. Their objective was to meticulously review all application materials and identify promising candidates to recommend to the HR manager.

A notable aspect of the scenario was the company's decision to incorporate AI in handling HR-related tasks. Participants, depending on the specific experimental condition they were in, were informed about the various roles AI would play in facilitating the recruitment process.

Below is a summary of our manipulation.

Assistive AI. Participants were presented with a general definition and explanation of assistive AI. In addition, participants were told that, in the recruitment task, AI would help them screen out unqualified applicants, organize candidates according to important filters, and direct to key content about candidates' backgrounds.

Augmented AI. Participants were informed of the general meaning of augmented AI. Next, participants were told that AI requires them to process information and make evaluations in a dramatically different way compared to traditional hiring processes. Precisely, AI can extract information from social media, infer their personality, and dig out "hidden messages" from reference letters. Hence, they need to utilize various forms of unconventional information when making decisions wisely.

Autonomous AI. Likewise, participants were first presented with a general introduction to autonomous AI and then its application in the recruitment task. They were told that they input the selection criteria into AI, and AI would organize and process applicants' information automatically and make recommendations about shortlisting candidates. As such, participants' role became the facilitator of AI performance with minimal decision-making power.

After reading the scenario, participants completed a brief questionnaire asking about their appraisal of the introduced AI and behavioral intentions.

7.3. Measures

All responses were recorded on a 7-point Likert scale, ranging from 1 (Strongly Disagree) to 7 (Strongly Agree).

Opportunity and threat appraisals. The items of opportunity were also adapted from Kraimer et al. (2005) and were adjusted to the scenario context. A sample item for opportunity appraisal is "To perform my job well, it is better for me to use AI". Items of threat appraisal were from McGregor and Elliot (2002) and adjusted for the AI adoption context. A sample item is "I viewed the adopted AI as a threat."

Intention of AI-learning behavior. We assessed AI-learning behavior based on Edmondson's (1999) 5-item scale. The items were modified to assess technology-related learning behavior. A sample item is "I will regularly take time to figure out ways to improve my technological ability".

Job insecurity. Job insecurity was assessed using the scale developed by Oldham et al. (1986). Sample items are "If this organization was facing economic problems, my job would be the first to go" and "I am insecure in my job".

Control variables. In this study, we controlled for gender and age. Gender was coded as 1 for male and 2 for female. We also controlled for participants' technology usage and knowledge about AI by asking them the percentage of total time spent on different types of electronic technologies and the extent they know what AI is (ranging from 1 "not at all" to 3 "very much"), as studies have found that technology familiarity can influence individuals' attitudes toward technology (Wejnert, 2002).

8. Study 2 Results

8.1. Test of main effects

Means, standard deviations, and intercorrelations are reported in Table 5.

Table 6 presents the regression results for the main effects of AI functions on appraisals, as well as the impact of these appraisals on work outcomes. After controlling for gender, age, technology usage, and AI knowledge, it was found that assistive AI was positively related to opportunity appraisal ($b = .25$, $p < .01$), and autonomous AI was positively related to threat appraisal ($b = .65$, $p < .05$). These findings support Hypotheses 1 and 2. However, augmented AI was negatively related to opportunity appraisal ($b = -.34$, $p < .05$) and shows no significant relation to threat appraisal ($b = -.17$, *n.s.*), thus, failing to support Hypothesis 3.

8.2. Test of mediations

Regression analyses revealed that opportunity appraisal was positively related to AI learning behavior ($b = .19$, $p < .01$) and negatively related to job insecurity ($b = -.40$, $p < .01$). Threat appraisal was not related to AI learning behavior ($b = -.02$, *n.s.*) and positive related to

Table 5
Descriptive Statistics and Bivariate Correlations (Study 2).

	Mean	SD	1	2	3	4	5	6	7	8	9	10
1.Gender	1.52	.50										
2.Age	37.24	10.23	−.04									
3.Technology Use	63.64	22.96	−.01	.00								
4.AI Knowledge	2.20	.48	−.29 * *	−.06	.00							
5.Assistive AI	.35	.48	−.05	−.06	.04	.00						
6.Augmented AI	.33	.47	.09	−.19 * *	−.01	−.01	−.51 * *					
7.Autonomous AI	.33	.47	−.03	.25 * *	−.04	.01	−.51 * *	−.49 * *				
8.Opportunity Appraisal	4.78	1.10	−.06	.01	−.05	.04	.24 * *	−.15 *	−.10			
9.Threat Appraisal	4.31	1.54	.05	.02	−.05	−.07	−.14 *	−.05	.19 * *	−.47 * *		
10.AI Learning Behavior	5.38	.84	.07	.13	.09	.08	.07	−.14 *	.07	.23 * *	−.04	
11.Job Insecurity	4.20	.98	.07	−.07	−.12	−.03	.00	−.08	.08	−.44 * *	.65 * *	−.19 * *

Note. $N = 217$. SD = standard deviation.
Gender: 1 = male, 2 = female.
Reliability coefficients appear along the diagonal.
* $p < .05$. * * $p < .01$.

Table 6
Summary of Regression Results (Study 2).

	Opportunity Appraisal		Opportunity Appraisal		Threat Appraisal		Threat Appraisal	
	b	*SE*	*b*	*SE*	*b*	*SE*	*b*	*SE*
Gender	−.04	.07	−.09	.15	.12	.22	.12	.21
Age	.02	.07	.00	.01	.00	.01	−.01	.01
Technology Use	−.06	.07	.00	.00	.00	.01	.00	.00
AI Knowledge	.03	.07	.06	.16	−.18	.23	−.20	.22
Assistive AI	.25 * *	.06						
Augmented AI			−.34 *	.16	−.17	.23		
Autonomous AI							.65 * *	.22
R^2	.07 *		.03		.01		.05	

Note. $N = 217$. SE = standard error.
Unstandardized regression coefficients are reported. * $p < .05$. * *$p < .01$.

job insecurity (b =.41, $p < .01$).

Based on these results, the Monte Carlo method with 20,000 replications were used to test the indirect effects of AI functions on work outcomes via appraisals (Preacher, Zyphur, and Zhang, 2010). Hypothesis 4 asserted that opportunity appraisal would mediate the relationships between assistive AI and AI learning behavior, and assistive AI and job insecurity. We found that the indirect effect of assistive AI on AI learning behavior via opportunity appraisal was.10 with a 95 % bias-corrected bootstrap CI of [.03,.20]. The indirect effect of assistive AI on job insecurity via opportunity appraisal was −.24 with a 95 % bias-corrected bootstrap CI of [-.38, −.11]. Thus, Hypothesis 4 was supported.

Hypothesis 5 predicted that opportunity appraisal would mediate the relationships between augmented AI and AI learning behavior, and between augmented AI and job insecurity. Not surprisingly, with the negative relationship between augmented AI and opportunity appraisal, the indirect effects of augmented AI on AI learning behavior (*Indirect effect* = −.06, 95 % CI [-.14, −.01]) and augmented AI on job insecurity (*Indirect effect* =.14, 95 % CI [.02,.28]) via opportunity appraisal were not as predicted, thus failing to support Hypothesis 5. Likewise, since the relationship between augmented AI and threat appraisal was not supported, the indirect effects of augmented AI on AI learning behavior (*Indirect effect* =.004, 95 % CI [-.02,.04]), and augmented AI on job insecurity (*Indirect effect* = −.07, 95 % CI [-.25,.10]), via threat appraisal were all not supported, thus failing to support Hypothesis 6.

Hypothesis 7a stated that threat appraisal would mediate the relationships between autonomous AI and AI learning behavior. The CI for the indirect effects of autonomous AI on AI learning behavior (*Indirect effect* = −.02, 95 % CI [-.08,.04]) included zero, failing to support Hypothesis 7a. In Hypothesis 7b, we predicted the indirect effect of autonomous AI on job insecurity via threat appraisal. The CI for this indirect effect (*Indirect effect* =.27, 95 % CI [.07,.48]) excluded zero, supporting Hypothesis 7b.

Study 2 supported the positive link between assistive AI and opportunity appraisal, the positive link between autonomous AI and threat appraisal, the indirect effects of assistive AI on AI learning behavior and job insecurity via opportunity appraisal, and the indirect effect of autonomous AI on job insecurity via threat appraisal. Overall, the hypothesized model received mixed support. To enhance the external validity of our findings, in Study 3, we sampled full-time employees whose jobs involved working with AI and tested the full theoretical model.

9. Study 3 method

9.1. Participants and procedure

The recruitment process was conducted in collaboration with a technology research company in China. Employees were invited to participate through direct invitations from the HR department. Prior to the study, participants were provided with detailed information regarding the study's objectives and the confidentiality measures in place to protect their data. Given the two-wave data collection design, employees were informed that to match their responses across both waves, they would be required to provide the last four digits of their phone numbers. This approach ensured anonymity while enabling response matching. Additionally, participants were notified that they would receive a monetary incentive of 75 RMB upon full completion of both survey waves.

This organization specializes in technical development, service provision, and technology transfer within the realms of new materials, new energy, and the Internet of Things (IoT). It also focuses on the integration of computer information systems. The company has been actively incorporating AI into its software development endeavors, primarily

focusing on enhancing the intelligence and adaptability of software solutions. A notable application of their AI technology is in streamlining the software testing processes. They have developed proprietary AI models capable of identifying potential defects, which contributes significantly to the overall improvement of software stability. Moreover, these AI tools are employed to scrutinize code repositories, offering recommendations to enhance code quality and even generating new code segments. Beyond software development, AI tools are also integral to project management within the company. These tools analyze various data points, monitor project progress, assess resource utilization, and identify potential risks.

The proposed model was tested in a two-wave study conducted over two weeks. The first wave focused on collecting data on three AI functions, AI self-efficacy, and demographic information. In the second wave, opportunity and threat appraisal, AI learning behavior, and job insecurity were measured. A total of 265 employees finished the first wave survey. The second survey was launched ten days later after the first one. Eventually, 237 participants completed the second survey, resulting in an 89 % response rate. After matching responses across the two waves, the final sample consisted of 222 participants, representing 83.8 % of the initial sample. Of these participants, 62.2 % were male, and 73.4 % held at least a bachelor's degree. The average age was 38.23 years (SD = 10.99). Additionally, 63.3 % had been employed at the company for at least one year, providing them with substantial workplace experience. 29.7 % of them held entry-level or junior positions, while only 4 % reported having little interest in new technologies or innovations.

9.2. Measures

All responses were recorded on a 7-point Likert scale, ranging from 1 (Strongly Disagree) to 7 (Strongly Agree). We followed the translation-back translation procedure to create the Chinese version of the measures. The coefficient alphas for assistive AI, augmented AI, autonomous AI, opportunity appraisal, threat appraisal, AI learning behavior, job insecurity, and AI self-efficacy were.92,.85,.94,.92,.95,.88,.92, and.71, respectively.

AI functions. AI functions were assessed with the measure validated in Study 1. Sample items are "The adopted AI is within the scope of my capabilities", "The adopted AI explores alternative options that I wouldn't have considered otherwise", and "The adopted AI can make decisions without my intervention" for assistive AI, augmented AI, and autonomous AI, respectively.

Opportunity and threat appraisal. The items for opportunity appraisal were sourced from Kraimer et al. (2005) and modified to suit the actual work context. A sample item for opportunity and threat appraisals is, "To perform my job well, it is better for me to use AI." The items for threat appraisal were adapted from McGregor and Elliot (2002) and adjusted to fit the AI context. One of the sample items is, "I viewed the adopted AI as a threat".

AI-learning behavior. As in study 2, AI-learning behavior was adapted from Edmondson's (1999) 5-item scale and modified to assess technology-related and actual learning behavior. A sample item is "I regularly take time to figure out ways to improve my technological ability".

Job insecurity. Job insecurity was assessed using the scale developed by Oldham et al. (1986). Sample items are "If this organization was facing economic problems, my job would be the first to go" and "I'm insecure in my job".

AI self-efficacy. The AI self-efficacy measure was adapted from the scale developed by Jones (1986) to reflect the AI context. A sample item is "I do not anticipate any problems in adjusting to AI adopted in my job".

Control variables. We controlled for gender, age, education, and job level.

10. Study 3 results

Table 7 presents the descriptive statistics and correlations among the study variables.

We used Mplus 7 to conduct a CFA and assess the distinctiveness of our key variables (assistive AI, augmented AI, autonomous AI, opportunity appraisal, threat appraisal, AI self-efficacy, AI learning behavior and job insecurity). As shown in Table 8, the proposed eight-factor model showed a good overall measurement fit with $\chi^2(637) = 1098.63$, CFI = .93, TLI = .92, SRMR = .06. All the factor loadings were significant, indicating convergent validity. We tested the discriminant validity of the proposed eight-factor model by contrasting it with alternative CFA models. The results presented in Table 8 substantiated discriminant validity.

10.1. Test of main effects

Table 9 shows the regression results. Assistive AI was positively related to opportunity appraisal (b =.39, p < .01); autonomous AI was positively related to threat appraisal (b =.18, p < .01); augmented AI was positively related to opportunity appraisal (b =.41, p < .01) but not significantly related to threat appraisal (b = −.16, *n.s.*). Therefore, Hypotheses 1 and 2 were supported and Hypothesis 3 was partially supported.

10.2. Test of mediations

Regression analyses indicated that opportunity appraisal was positively related to AI learning behavior (b =.34, p < .01) and negatively related to job insecurity (b = −.35, p < .01), while threat appraisal was negatively related to AI learning behavior (b =.12, p < .01) and positively related to job insecurity (b =.20, p < .01).

As in Study 2, the Monte Carlo method was utilized to test indirect effects. Supporting Hypotheses 4a and 4b, assistive AI was found to have indirect effects on AI learning behavior (*Indirect effect* =.15, 95 % CI [.02,.27]) and job insecurity (*Indirect effect* = −.09, 95 % CI [-.18, −.001]) via opportunity appraisal. In support of Hypotheses 5a and 5b, the indirect effects of augmented AI on AI learning behavior (*Indirect effect* =.12, 95 % CI [.06,.21]) and job insecurity (*Indirect effect* = −.09, 95 % CI [-.19, −.003]) through opportunity appraisal were all significant, with confidence intervals not including zero.

Hypotheses 6a and 6b posited that threat appraisal would mediate the relationships between augmented AI and AI learning behavior, as well as job insecurity. However, the indirect effects of augmented AI on AI learning behavior (*Indirect effect* =.05, 95 % CI [-.07,.17]) and on job insecurity (*Indirect effect* = −.03, 95 % CI [-.09, −.01]) were not supported, thereby failing to confirm Hypotheses 6a and 6b.

Hypotheses 7a and 7b predicted that threat appraisal would mediate the relationships between autonomous AI and outcomes. Results showed that the indirect effect of autonomous AI on AI learning behavior was −.02 with a 95 % CI of [-.05, −.001], supporting Hypothesis 7a. In addition, the indirect effect of autonomous AI on job insecurity via threat appraisal excluded zero (*Indirect effect* =.03, 95 % CI [-.07, −.004]), supporting Hypothesis 7b.

10.3. Test of moderations and moderated mediations

Hypotheses 8 and 10 focus on the moderating influence of AI self-efficacy in the dynamics between augmented AI and autonomous AI, and their respective impacts on threat appraisal. The data presented in Table 10, specifically in Models 5 and 6, indicate that the anticipated moderating effect of AI self-efficacy on the relationships between both augmented AI and threat appraisal, as well as autonomous AI and threat appraisal, was not observed. Hypotheses 8 and 10 were thus not supported.

Hypotheses 9 and 11 focused on examining the moderating effects of

Table 7
Descriptive Statistics and Bivariate Correlations (Study 3).

	M	SD	1	2	3	4	5	6	7	8	9	10	11
1. Gender	1.38	.49											
2. Age	38.23	10.99	.06										
3. Education	4.25	1.40	−.07	−.03									
4. Job level	3.29	1.15	−.12	.30 * *	.26 * *								
5. Assistive AI	5.47	1.08	.07	−.03	.01	.14 *							
6. Augmented AI	4.71	1.10	−.06	.09	.02	.21 * *	.26 * *						
7. Autonomous AI	4.27	1.61	.01	.12	−.18 * *	.05	.11	.39 * *					
8. Opportunity appraisal	5.57	1.07	−.12	.02	−.01	.13	.50 * *	.54 * *	.26 * *				
9. Threat appraisal	2.60	1.55	−.14 *	−.01	.02	.02	−.12	−.05	.13	−.32 * *			
10. Learning behaviour	5.60	.94	−.06	.03	.11	.17 *	.33 * *	.18 * *	.06	.40 * *	−.18 * *		
11. Job insecurity	5.30	1.17	.01	−.15 *	.06	.12	.33 * *	.21 * *	.06	.33 * *	−.26 * *	.37 * *	
12. AI Self-efficacy	5.29	.88	.04	−.05	−.06	.07	.46 * *	.22 * *	.12	.49 * *	−.25 * *	.52 * *	.43 * *

Note. $N = 222$. *SD* = standard deviation.
Gender: 1 = male, 2 = female.
Reliability coefficients appear along the diagonal.
*$P < .05$. * *$P < .01$.

Table 8
CFA Results (Study 3).

Model	χ^2	df	CFI	TLI	SMER
Eight-factor model	1098.63	637	.93	.92	.06
Five-factor model 1[a]	2719.26	655	.66	.64	.14
Five-factor model 2[b]	3270.02	664	.58	.55	.16
One-factor model[c]	4792.84	665	.33	.29	.15

Note. $N = 222$.
[a] This model combines, from the seven-factor model, assistive AI, augmented AI, autonomous AI, and AI self-efficacy to form a Time 1 factor.
[b] This model combines, from the seven-factor model, opportunity appraisal, threat appraisal, AI learning behavior and job insecurity to form a Time 2 factor.
[c] We combined all measurement items into one grand factor.

AI self-efficacy on the indirect relationships between augmented AI and autonomous AI, and work outcomes, mediated by threat appraisal. We calculated the indirect effects at both higher (+1 SD) and lower (-1 SD) levels of AI self-efficacy. The findings are presented in Table 10. Contrary to our expectations, Hypothesis 9 did not receive empirical support. This outcome aligns with the previous observation that the anticipated negative relationship between augmented AI and threat appraisal was not empirically validated.

Hypothesis 11 stated that AI self-efficacy would moderate the indirect relationships between autonomous AI and work outcomes through threat appraisal. Although the moderating effect of AI self-efficacy on the negative relationship between autonomous AI and threat appraisal was not supported, the results showed that the indirect effect of autonomous AI on AI learning behavior (*Indirect effect* = −0.02, 95 % CI [-0.05, −0.003]) and job insecurity (*Indirect effect* = 0.04, 95 % CI [0.01, 0.08]) via threat appraisal was significant only when AI self-efficacy was low, rather than high. Therefore, Hypothesis 11 was partially supported.

11. General discussion

The current research sheds light on how perceived AI functions, through psychological appraisals, shape employees' attitudes and behaviors toward AI adoption. Our findings suggest that assistive AI leads to opportunity appraisal, which in turn heightens AI learning behavior and reduces job insecurity, while autonomous AI leads to threat

Table 9
Summary of Regression Results (Study 3).

	Opportunity Appraisal		Threat Appraisal		Threat Appraisal		Threat Appraisal	
	Model 1		Model 3		Model 4		Model 5	
	b	*SE*	*b*	*SE*	*b*	*SE*	*b*	*SE*
Gender	−.28 *	.11	−.44 *	.22	−.42 *	.21	−.41	.21
Age	.00	.01	.00	.01	−.00	.01	−.00	.01
Education	−.01	.04	.04	.08	.02	.08	.01	.08
Job level	−.03	.05	.04	.10	.05	.10	.06	.10
Assistive AI	.39 * *	.05	−.15	.10	−.00	.11	−.02	.11
Augmented AI	.41 * *	.06	−.16	.11	.01	.49	−.13	.10
Autonomous AI	.04	.04	.18 *	.07	.19 * *	.07	−.31	.35
AI Self-efficacy					.01	.43	−.81 * *	.30
Augmented AI x Self-efficacy					−.02	.09		
Autonomous AI x Self-efficacy							.09	.06
R^2	.45 * *		.06 * *		.11 * *		.12 * *	

Note. $N = 222$. *SE* = standard error.
Unstandardized regression coefficients are reported. * $p < .05$. * *$p < .01$.

Table 10
Moderated Mediation Results (Study 3).

Moderator: AI Self-efficacy				95 % CI	
		Indirect Effect	*SE*	Lower Limit	Upper Limit
Augmented AI-Threat Appraisal-AI Learning Behavior	Index of Moderated Mediation	.002	.01	−.03	.03
	Low AI Self-efficacy (−1 SD)	01	.02	−.01	.04
	High AI Self-efficacy (+1 SD)	.01	.01	−.01	.05
Augmented AI-Threat Appraisal-Job Insecurity	Index of Moderated Mediation	−.004	.02	−.05	.05
	Low AI Self-efficacy (−1 SD)	−.02	.04	−.10	.04
	High AI Self-efficacy (+1 SD)	−.02	.02	−.08	.01
Autonomous AI-Threat Appraisal-AI Learning Behavior	Index of Moderated Mediation	−.01	.01	−.02	.01
	Low AI Self-efficacy (−1 SD)	**−.02**	**.01**	**−.05**	**−.003**
	High AI Self-efficacy (+1 SD)	−.01	.02	−.05	.01
Autonomous AI-Threat Appraisal-Job Insecurity	Index of Moderated Mediation	.02	.02	−.02	.04
	Low AI Self-efficacy (−1 SD)	**.04**	**.02**	**.01**	**.08**
	High AI Self-efficacy (+1 SD)	.02	.02	−.02	.08

Note. $N = 222$. SE = standard error.
Unstandardized regression coefficients are reported. * $p < .05$. * *$p < .01$

appraisal and in turn increases job insecurity. Results concerning the impact of augmented AI were inconsistent across the three studies. Moreover, the buffering role of AI self-efficacy in the impact of augmented AI and autonomous AI on threat appraisal received no support (Table 11 summarize the design and results of the three studies). Overall, employees' reactions to AI depend largely on perceived AI functions, and the impact of augmented AI on employees remains inconclusive.

12. Interpretation of research findings

Each of these AI functions produced distinct outcomes, underscoring the necessity for a discerning approach to differentiate between them. Our studies revealed that when AI was perceived with assistive and augmented capacities, both enabling employees to maintain control over tasks that incorporate AI, it fostered opportunity appraisal, which was strongly associated with an increase in engagement in AI-related learning activities and a decrease in job insecurity. In contrast, perceiving AI as dominating task control, thereby relegating employees to mere executors of AI-generated directives, tends to amplify job insecurity and diminish learning engagement. The commonality between assistive and augmented AI is that they both allow employees to retain control over their work. This contrasts with autonomous AI, where control shifts from the employees to AI. These findings highlight the importance of control in task execution, aligning with previous research indicating that employees prefer having control and autonomy in their work (e.g., Bond and Bunce, 2003). Moreover, when AI self-efficacy was low, autonomous AI, via threat appraisal, had significant indirect effects on AI learning behavior and job insecurity. These results suggest, those who are not confident with their AI abilities may be more negatively influenced by autonomous AI.

The results concerning augmented AI, when compared to assistive AI and autonomous AI, displayed inconsistency across the three studies. Specifically, in Studies 1 and 3, augmented AI showed a positive correlation with opportunity appraisal, yet no significant relationship with threat appraisal was observed. Conversely, Study 2 demonstrated a negative association with opportunity appraisal and no correlation with threat appraisal. These divergent outcomes suggest that employees may perceive augmented AI as more ambiguous compared to assistive and autonomous AI. This ambiguity could be attributed to whether employees' actual work environments have integrated AI technologies. In Studies 1 and 3, participants were specifically chosen from workplaces that had implemented AI technologies. In contrast, Study 2 did not have this prerequisite; participants were presented with a hypothetical scenario illustrating AI's role in "their job". This difference in experiential background could account for the varied responses. Participants in Study 2, who may not have direct experience with AI in their work, likely lacked a practical understanding of AI's applications and its implications. On the other hand, employees with direct exposure to AI in their workplaces could form a more practical and comprehensive perception of its functionality and influence—although augmented AI is capable of performing tasks beyond human capabilities, as long as they still maintain the control over the tasks, they tend to form opportunity perception towards AI applications.

12.1. Theoretical implications

First and foremost, our study enhances the understanding of employee psychological and behavioral responses to the introduction of AI in the workplace. Previous research attempting to gauge AI's impact on employees has often treated AI as uniform, failing to consider its varied functions. This approach has led to an overly simplistic and polarized perspective, concluding AI's potential positive or negative influence on employees without acknowledging its multifaceted nature (Carter and Nielsen, 2017; Korinek and Stiglitz, 2018). This research builds upon existing literature by differentiating between the roles of AI and employees in terms of agency. It systematically examines the various functions of AI that elicit positive responses from employees, as well as those that trigger negative reactions. Additionally, specific individual characteristics are explored, with a particular focus on AI self-efficacy, to understand how this trait can mitigate potential adverse effects associated with AI integration in the workplace. Through this study, we aim to provide a comprehensive and nuanced understanding of why AI can lead to divergent outcomes in the workplace.

Second, by systematically examining how different AI functions influence employees' cognitive appraisals, this study extends prior research on technology adoption and workplace behavior. Traditional models, such as the TAM (Davis, 1989), primarily focus on AI adoption in terms of ease of use and perceived usefulness but do not fully capture the psychological mechanisms through which AI adoption influences employees. This study builds upon CAT (Lazarus and Folkman, 1984) and threat-rigidity theory (Staw et al., 1981) to explain how employees interpret AI's role in their work and how these interpretations shape their learning behavior and job insecurity. A key theoretical insight from this study is that the target of cognitive appraisal may not necessarily be an objective entity. Instead, individuals may first develop distinctive subjective perceptions of AI functions before forming appraisals based on these perceptions.

Lastly, the development and validation of the AI functions measure hold important implications for future research, as they address the need for a practical approach to measuring AI functions, a necessity highlighted in previous studies (Duan et al., 2019). Most current studies are either conceptual analyses exploring the potential impact of AI

Table 11
Study Summary Table.

Study	Sample	Study Design	Key Variables	Analytic Method	Key Findings
Study 1a: Item Generation	19 Ph.D. students	Expert evaluation of AI function items	Assistive, Augmented, and Autonomous AI items	Content Validity Index (CVI)	Initial pool of AI function items
Study 1b: Exploratory Factor Analysis (EFA)	176 AI-related professionals from Prolific	Survey-based factor analysis	The three AI function dimensions	Exploratory Factor Analysis (EFA)	Identified three-factor structure
Study 1c: Confirmatory Factor Analysis (CFA)	200 AI-related professionals from Prolific	Survey-based confirmatory factor analysis	The three AI function dimensions	Confirmatory Factor Analysis (CFA)	Confirmed three-dimensional model
Study 1d: Validity Testing	200 AI-related professionals from Prolific	Survey-based discriminant and criterion validity testing	The three AI functions, AI familiarity, AI attachment, trust stance in AI, opportunity appraisal, threat appraisal	Discriminant validity testRegression analysis	Confirmed discriminant validityAI functions predict opportunity and threat appraisals
Study 2	218 full-time employees from Prolific	Experimental study	The three AI functions, opportunity appraisal, threat appraisal, AI learning behavior, job insecurity	Regression analysis to test main effects of AI functions on appraisals and work outcomes. Monte Carlo method to examine indirect effects of AI functions via appraisals.	Assistive AI increased opportunity appraisal, promoting AI learning, and reducing job insecurity.Autonomous AI increased job insecurity via threat appraisal but did not affect AI learning. Unexpectedly, Augmented AI reduced opportunity appraisal and did not significantly trigger threat appraisal.
Study 3	222 employees from a technology company in China	Two-wave longitudinal field study	The three AI functions, AI self-efficacy, opportunity appraisal, threat appraisal, AI learning behavior, job insecurity	CFA for measurement validationRegression analysis for main effectsMonte Carlo method for indirect effectsMplus for moderation analyses	Assistive AI increased opportunity appraisal, boosting AI learning and reducing job insecurity.Autonomous AI was perceived as a threat, leading to higher job insecurity and lower AI learning. Augmented AI was seen as beneficial, promoting AI learning and reducing job insecurity (contrary to Study 2).AI self-efficacy mitigated the negative effects of autonomous AI on job insecurity but had no significant impact on augmented AI.

(Chowdhury et al., 2022; Duan et al., 2019), rely on experimental methodologies (Leung et al., 2018; Tong et al., 2021), or do not directly engage with AI characteristics in their measurements (Pan et al., 2023). This results in a lack of empirical evidence, particularly in field settings, regarding how, why, and to what extent AI impacts the workforce. The findings of this research, especially Study 3, which was conducted in an organization that truly adopted AI, demonstrate the reliability and validity of the assistive-augmented-autonomous AI functions measure. This measure may aid future researchers in examining AI in field settings and in enhancing external validity.

12.2. Practical implications

Many organizations have adopted or are planning to adopt AI in order to increase productivity and thrive in a competitive economy. Our findings provide unique insights for such organizations from the perspective of how their employees may react psychologically and behaviorally to AI. From the organization's perspective, our research demonstrates the need for organizations to understand how employees appraise AI and how they should provide employees with adequate resources, such as instrumental knowledge to help with their learning behavior or social support to ease employees' job insecurity so that they can adapt better to AI-intruded work environments. In particular, considering that autonomous AI is likely to elicit threat appraisal and further lead to reduced learning behavior and increased job insecurity, organizations should be cautious when introducing AI into the workplace and communicate its functions to employees, especially when AI can make decisions without human intervention. Otherwise, employees' job insecurity and reduced learning behavior may backfire on organizational performance.

Practically, organizations can embed open innovation principles into AI adoption to enhance the synergy between AI and employee expertise and reduce employee negative reactions. To be specific, organizations can cultivate an open innovation culture that promotes transparency and inclusivity in AI adoption by integrating AI into open innovation frameworks that emphasize collaboration between human employees and intelligent systems. In addition, organizations can create opportunities for cross-functional collaboration through initiatives like AI-focused hackathons or workshops, where employees from diverse departments to share best practices, exchange experiences and learn about the latest advancements.

From the perspective of employees, the assistive-augmented-autonomous categorization reflects how one construes AI functions in the workplace. Although AI functions can be partially determined by objective characteristics, there is much subjectivity in perceiving AI. Upon facing the same AI application, some employees may perceive it as assistive, while others may view it as augmented or even autonomous. The distinctive views about AI may depend on individual characteristics. Employees may benefit from learning AI-related knowledge and cultivating innovation mindsets, so that they can develop competitive advantages in a reformed work environment with AI. Furthermore, employees may enhance their AI self-efficacy in order to control their negative attitudes and become effective collaborators of AI.

13. Limitations and future directions

Despite validating AI functions and testing hypotheses using diverse samples and methodologies, our research has several limitations. First, methodological concerns should be acknowledged. In Study 2, the scenario-based experimental approach lacked real interactions between humans and AI, particularly in the augmented AI condition, which entails a more complex and ambiguous nature. Additionally, while Study 2 employed a randomized experimental design to establish causal relationships between AI functions and appraisals, the simultaneous

collection of appraisals and outcomes raises concerns about potential common method variance. To address these limitations, Study 1 and Study 3 incorporated full-time employees whose work involves AI, providing insights based on actual workplace experiences. However, Study 1 relied on participants recruited through an online panel. Online panel participants tend to be more technologically literate (AnswerLab, 2023), which may have led to a sample skewed toward individuals with higher technological literacy, limiting the generalizability of findings to employees with lower AI exposure. Furthermore, as Study 1 collected self-reported measures at a single point in time, it does not capture how AI-related perceptions evolve over time. Study 3 mitigated this limitation to some extent by employing a two-wave data collection design. However, the reliance on self-reported measures still introduces the potential for common method variance. While we attempted to reduce this issue by implementing a two-wave data collection strategy, future research should consider integrating multiple data sources, such as supervisor or peer evaluations or behavioral tracking, to enhance the robustness of findings and minimize self-report biases.

Second, this research is grounded in a cognitive appraisal perspective, focusing on how AI functions influence work outcomes through psychological appraisals. While CAT offers valuable insights into how employees assess AI adoption, it has certain limitations. It primarily explains immediate reactions to AI but does not fully capture long-term adaptation processes. Additionally, CAT focuses on individual cognition and does not explicitly consider broader organizational factors, such as leadership or workplace culture, which may also shape employees' responses to AI. To address these limitations, this study incorporates threat-rigidity theory to explain how individual appraisals translate into specific workplace behaviors, thereby offering a more comprehensive framework. However, future research should move beyond an exclusive focus on individual cognition and consider contextual factors, such as leadership and organizational support, to develop a more holistic understanding of employees' responses to AI adoption.

Third, AI self-efficacy serves as an important moderating factor, building on the well-established concept that self-efficacy is a key coping resource (Schwarzer et al., 2005). Our research has revealed that when AI self-efficacy is low, as opposed to high, the indirect impact of autonomous AI on learning behavior and job insecurity, mediated through threat appraisal, becomes significant. This finding highlights the importance of self-efficacy in moderating the adverse effects of autonomous AI. Consequently, we advocate for future research to delve deeper into various factors, such as AI knowledge and experience, that could influence AI self-efficacy and mitigate the adverse impacts of autonomous AI.

14. Conclusion

The present study proposes and validates a typology of three AI functions: assistive, augmented, and autonomous AI. Additionally, employees' attitudes and behavioral responses to AI adoption in the workplace are examined. The findings indicate that the nature of employees' reactions—whether positive or negative—to AI adoption is largely shaped by the specific perceived roles that AI assumes in their work tasks. The insights gained from this research contribute to a deeper understanding of how the workforce adapts to an evolving work environment driven by the increasing adoption of AI.(Fig. 1)

Funding acknowledgements

This research was supported by the Start-up Grant (RF10018N) from Singapore's Ministry of Education (MOE), obtained by Jieqiong. Ng Teng Fong Charitable Foundation (R-317-000-156-720). The funding provided financial support for data collection. The sponsor had no role in the study design, data analysis, interpretation of results, writing of the manuscript, or the decision to submit this article for publication.

Author note

This study is supported by the Start-up Grant (RF10018N) from Singapore's Ministry of Education (MOE).

Ethical statement

This manuscript adheres to the reporting standards as outlined by the journal. All data collection and retention processes were conducted in strict compliance with the requirements of the SUSS Institutional Review Board (IRB). In accordance with SUSS IRB guidelines, the collected data cannot be made publicly available.

The content of this manuscript is entirely original. The authors confirm that the manuscript has not been previously published, nor is it under consideration for publication elsewhere. Proper citations have been provided for all referenced works, ensuring that all sources are appropriately acknowledged.

Authorship of this paper has been determined based on contributions. The first author was responsible for the majority of the work, including manuscript writing, data collection, and data analysis. The second and third authors contributed to the revision of the manuscript, while the fourth author supervised and ensured the overall quality of the work. The fifth author participated in data collection and analysis.

The author team affirms that no generative AI was used to generate content for this manuscript. However, AI-assisted tools were utilized solely for language refinement and grammar checking under human oversight.

This research did not involve any procedures that could harm human or animal subjects. All ethical considerations have been carefully adhered to throughout the research process.

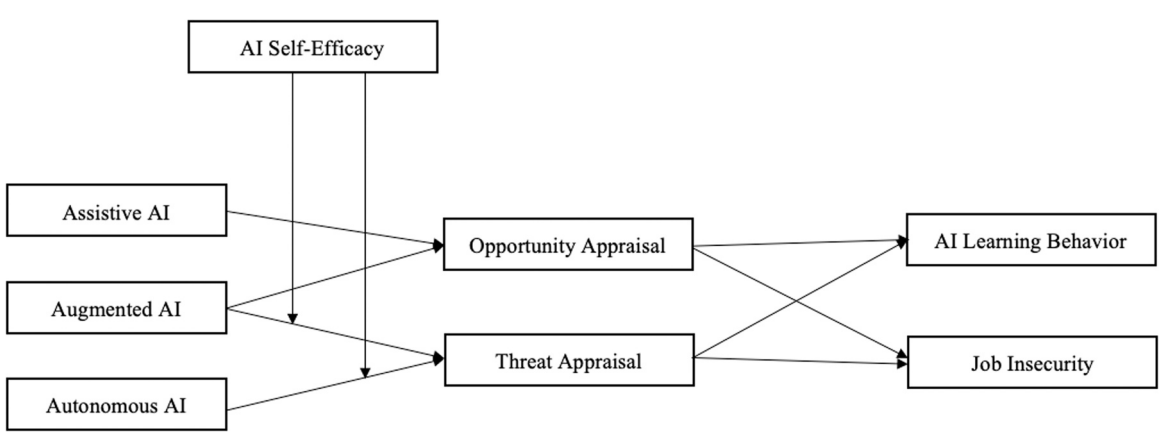

Fig. 1. Proposed Model.

CRediT authorship contribution statement

Zhang Fengzhi: Resources, Methodology, Investigation, Formal analysis, Data curation. **Song Zhaoli:** Supervision, Resources, Funding acquisition, Conceptualization. **Cao Jieqiong:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yao Jingxian:** Writing – review & editing, Methodology, Conceptualization. **Sun Shuhua:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Variable Measurement Scale

Assistive AI

- The adopted AI is within the scope of my capabilities.
- The tasks performed by AI are within my control.
- The adopted AI operates within the boundaries of my skill set.
- The adopted AI operates within the range of my abilities and expertise.
- The adopted AI performs tasks within my capabilities

Augmented AI

- The adopted AI explores alternative options that I wouldn't have considered otherwise.
- The way the adopted AI performs work tasks is beyond what I can do.
- The adopted AI provides new approaches to work tasks with its advanced capabilities.
- The adopted AI offers me recommendations based on its findings which I am unable to generate by myself.
- The adopted AI expands my thoughts beyond my current knowledge base.

Autonomous AI

- The adopted AI can make decisions without my intervention.
- The adopted AI can take full control of specific tasks without human intervention.
- The adopted AI can make decisions in real-time, navigating complex situations with minimal or no human input.
- The adopted AI exercises decision-making capabilities independently of my input.
- The adopted AI can make its own decisions in complex environments.

Opportunity Appraisal (Kraimer, Wayne, Liden, and Sparrowe, 2005)

- To perform my job well, it would be better for me to use AI.
- AI will help me to better meet my job demands.
- It would be best for me if my organization use AI.
- It would be best for me if AI is adopted to my job.
- There are a number of benefits associated with the use of AI.

Threat Appraisal (McGregor and Elliot, 2002)

- I viewed the adopted AI as a threat.
- The adopted AI presented itself as a threat to me.
- I felt threatened by the adopted AI.

AI Learning behavior (Edmondson, 1999)

- I will regularly take time to figure out ways to improve my technological ability.
- I will go out and get all the technology information I possibly can from others – such as customers, or other parts of the organization.
- I will frequently seek new information that leads me to better understand technology.
- I will always make sure that I stop to reflect on the technological work process.
- I will speak up to test assumptions about technology issues under discussion.

Job insecurity (Oldham et al., 1986)

- I'll be able to keep my present job as long as I wish.
- My organization will not cut back on the number of hours I work each week.
- If this organization were facing economic problems, my job would be the first to go.
- I am confident that I will be able to work for this organization as long as I wish.
- My job will be there as long as I want it.
- If my job were eliminated, I would be offered another job in the organization.
- Regardless of economic conditions, I will have a job at my current organization.
- I am secure in my job.
- The organization would transfer me to another job if I were laid off from my present job.
- My job is not a secure one.

AI self-efficacy (Jones, 1986)

- With the use of AI, my job is well within the scope of my abilities.
- I do not anticipate any problems in adjusting to AI adopted in my job.
- I feel I am overqualified for work tasks involving AI.
- I have all the technical knowledge I need to deal with AI, all I need now is practical experience.
- I feel confident that my skills and abilities regarding AI equal or exceed those of my future colleagues.
- My past experiences and accomplishments increase my confidence that I will be able to perform successfully for work tasks involving AI.
- With the use of AI, I will be able to handle a more challenging job than the one I will be doing.
- If AI is adopted, my job will exactly satisfy my expectations of myself.

References

Almanasreh, E., Moles, R., Chen, T.F., 2019. Evaluation of methods used for estimating content validity. Res. Soc. Adm. Pharm. 15 (2), 214–221. https://doi.org/10.1016/j.sapharm.2018.03.066.

AnswerLab, 2023. How to recruit participants with low tech skills for UX research. AnswerLab. 〈https://www.answerlab.com/insights/low-tech-user-research-ops〉.

Aquino, K., Douglas, S., 2003. Identity threat and antisocial behavior in organizations: the moderating effects of individual differences, aggressive modeling, and hierarchical status. Organ. Behav. Hum. Decis. Process. 90, 195–208. https://doi.org/10.1016/s0749-5978(02)00517-4.

Argyris, C., Schön, D.A., 1997. Organizational learning: a theory of action perspective. Reis 77/78, 345–348.

Ashford, S.J., Lee, C., Bobko, P., 1989. Content, cause, and consequences of job insecurity: a theory-based measure and substantive test. Acad. Manag. J. 32 (4), 803–829. https://doi.org/10.5465/256569.

Baird, A., Maruping, L.M., 2021. The next generation of research on IS use: a theoretical framework of delegation to and from agentic IS artifacts. MIS Q. 45 (1), 315–341. https://doi.org/10.25300/misq/2021/15882.

Baldwin, T.T., Magjuka, R.J., 2014. Training as an organizational episode: Pretraining influences on trainee motivation. In *Improving training effectiveness in work organizations*. Psychology Press.
Bandura, A., 1997. Self-efficacy: The exercise of control. Freeman, New York.
Barr, P.S., Glynn, M.A., 2004. Cultural variations in strategic issue interpretation: relating cultural uncertainty avoidance to controllability in discriminating threat and opportunity. Strateg. Manag. J. 25, 59–67. https://doi.org/10.1002/smj.361.
Beaudry, A., Pinsonneault, A., 2005. Understanding user responses to information technology: a coping model of user adaptation. MIS Q. 29 (3), 493–524. https://doi.org/10.2307/25148693.
Bond, F.W., Bunce, D., 2003. The role of acceptance and job control in mental health, job satisfaction, and work performance. J. Appl. Psychol. 88 (6), 1057–1067. https://doi.org/10.1037/0021-9010.88.6.1057.
Bosma, H., Stansfeld, S.A., Marmot, M.G., 1998. Job control, personal characteristics, and heart disease. J. Occup. Health Psychol. 3, 402–409. https://doi.org/10.1037/1076-8998.3.4.402.
Britton, J.C., Lissek, S., Grillon, C., Norcross, M.A., Pine, D.S., 2011. Development of anxiety: the role of threat appraisal and fear learning. Depress Anxiety 28 (1), 5–17. https://doi.org/10.1002/da.20733.
Broekhuizen, T., Dekker, H., de Faria, P., Firk, S., Nguyen, D.K., Sofka, W., 2023. AI for managing open innovation: opportunities, challenges, and a research agenda. J. Bus. Res. 167, 114196. https://doi.org/10.1016/j.jbusres.2023.114196.
Brown, S.P., Ganesan, S., Challagalla, G., 2001. Self-efficacy as a moderator of information-seeking effectiveness. J. Appl. Psychol. 86, 1043–1051. https://doi.org/10.1037/0021-9010.86.5.1043.
Carter, S., Nielsen, M., 2017. Using artificial intelligence to augment human intelligence. Distill 2 (12), e9. https://doi.org/10.23915/distill.00009.
Castelo, N., 2019. Blurring the line between human and machine: Marketing artificial intelligence (doctoral dissertation). Retrieved from. Columbia University Academic Commons. https://doi.org/10.7916/d8-k7vk-0s40.
Čerka, P., Grigienė, J., Sirbikytė, G., 2015. Liability for damages caused by artificial intelligence. Comput. Law Secur. Rev. 31 (3), 376–389. https://doi.org/10.1016/j.clsr.2015.03.008.
Chemers, M.M., Watson, C.B., May, S.T., 2000. Dispositional affect and leadership effectiveness: a comparison of self-esteem, optimism, and efficacy. Personal. Soc. Psychol. Bull. 26, 267–277. https://doi.org/10.1177/0146167200265001.
Chi, O.H., Jia, S., Li, Y., Gursoy, D., 2021. Developing a formative scale to measure consumers' trust toward interaction with artificially intelligent (AI) social robots in service delivery. Comput. Hum. Behav. 118 (2021), 1–17. https://doi.org/10.1016/j.chb.2021.106700.
Chowdhury, S., Budhwar, P., Dey, P.K., Joel-Edgar, S., Abadie, A., 2022. AI-employee collaboration and business performance: integrating knowledge-based view, socio-technical systems and organisational socialisation framework. J. Bus. Res. 144, 31–49. https://doi.org/10.1016/j.jbusres.2022.01.069.
Chui, M., Harryson, M., Manyika, J., Roberts, R., Chung, R., van Heteren, A., Nel, P., 2018. Notes from the AI frontier: Applying AI for social good. McKinsey Global Institute. ⟨https://www.mckinsey.com/~/media/mckinsey/featured%20insights/artificial%20intelligence/applying%20artificial%20intelligence%20for%20social%20good/mgi-applying-ai-for-social-good-discussion-paper-dec-2018.pdf⟩.
D'Souza, R.M., Strazdins, L., Lim, L.L., Broom, D.H., Rodgers, B., 2003. Work and health in a contemporary society: demands, control, and insecurity. J. Epidemiol. Community Health 57 (11), 849–854. https://doi.org/10.1136/jech.57.11.849.
Daugherty, P.R., Wilson, H.J., Chowdhury, R., 2019. Using artificial intelligence to promote diversity. MIT Sloan Manag. Rev. ⟨https://diversityatlas.io/wp-content/uploads/2023/08/2019-Using-Artificial-Intelligence-to-Promote-Diversity.pdf⟩.
Davenport, T.H., Bean, R., 2018. Big companies are embracing analytics, but most still don't have a data-driven culture. Harv. Bus. Rev. 6, 1–4.
Davis, F.D., Bagozzi, R.P., Warshaw, P.R., 1989. User acceptance of computer technology: a comparison of two theoretical models. Manag. Sci. 35 (8), 982–1003. https://doi.org/10.1287/mnsc.35.8.982.
DHL Group, 2024. DHL Supply Chain implements generative AI to enhance data management, customer support, and proposal accuracy. DHL Press Release. ⟨https://group.dhl.com/en/media-relations/press-releases/2024/dhl-supply-chain-implements-generative-ai.html⟩.
Dietvorst, B.J., Simmons, J.P., Massey, C., 2016. Overcoming algorithm aversion: people will use imperfect algorithms if they can (even slightly) modify them. Manag. Sci. 64, 1155–1170. https://doi.org/10.1287/mnsc.2016.2643.
Duan, Y., Edwards, J.S., Dwivedi, Y.K., 2019. Artificial intelligence for decision making in the era of Big Data–evolution, challenges and research agenda. Int. J. Inf. Manag. 48, 63–71. https://doi.org/10.1016/j.ijinfomgt.2019.01.021.
Dutton, J.E., Jackson, S.E., 1987. Categorizing strategic issues: links to organizational action. Acad. Manag. Rev. 12, 76–90. https://doi.org/10.5465/amr.1987.4306483.
Edmondson, A., 1999. Psychological safety and learning behavior in work teams. Adm. Sci. Q. 44, 350–383. https://doi.org/10.2307/2666999.
Fisher, C.D., Minbashian, A., Beckmann, N., Wood, R.E., 2013. Task appraisals, emotions, and performance goal orientation. J. Appl. Psychol. 98, 364. https://doi.org/10.1037/a0031260.
Folkman, S., Lazarus, R.S., 1985. If it changes it must be a process: study of emotion and coping during three stages of a college examination. J. Personal. Soc. Psychol. 48 (1), 150–170.
Glikson, E., Woolley, A.W., 2020. Human trust in artificial intelligence: review of empirical research. Acad. Manag. Ann. 14 (2), 627–660. https://doi.org/10.5465/annals.2018.0057.
Greenhalgh, L., Rosenblatt, Z., 1984. Job insecurity: toward conceptual clarity. Acad. Manag. Rev. 9, 438–448. https://doi.org/10.5465/amr.1984.4279673.
Hackman, J.R., Oldham, G.R., 1975. Development of the job diagnostic survey. J. Appl. Psychol. 60, 159–170. https://doi.org/10.1037/h0076546.
Haenlein, M., Kaplan, A., 2019. A brief history of artificial intelligence: on the past, present, and future of artificial intelligence. Calif. Manag. Rev. 61 (4), 5–14. https://doi.org/10.1177/0008125619864925.
Heuven, E., Bakker, A.B., Schaufeli, W.B., Huisman, N., 2006. The role of self-efficacy in performing emotion work. J. Vocat. Behav. 69, 222–235. https://doi.org/10.1016/j.jvb.2006.03.002.
Hinkin, T.R., 1995. A review of scale development practices in the study of organizations. J. Manag. 21 (5), 967–988.
Hsieh, P., Sullivan, J.R., Guerra, N.S., 2007. A closer look at college students: self-efficacy and goal orientation. J. Adv. Acad. 18, 454–476. https://doi.org/10.4219/jaa-2007-500.
Hu, L.T., Bentler, P.M., 1999. Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. Structural equation modeling: a multidisciplinary journal 6 (1), 1–55. https://doi.org/10.1080/10705519909540118.
Jaiswal, A., Arun, C.J., Varma, A., 2022. Rebooting employees: upskilling for artificial intelligence in multinational corporations. Int. J. Hum. Resour. Manag. 33 (6), 1179–1208. https://doi.org/10.1080/09585192.2021.1891114.
Jalil, A., 2024. Almost all Co. Will. Use AI Financ. Report. 3 years: KPMG. ⟨https://www.nst.com.my/business/corporate/2024/07/1078003/almost-all-companies-will-use-ai-financial-reporting-3-years-kpmg#google_vignette⟩.
Jarrahi, M.H., 2018. Artificial intelligence and the future of work: human-AI symbiosis in organizational decision making. Bus. Horiz. 61, 577–586. https://doi.org/10.1016/j.bushor.2018.03.007.
Jerusalem, M., Schwarzer, R., 1992. Self-efficacy as a resource factor in stress appraisal processes. In: Schwarzer, R. (Ed.), Self-efficacy: Thought control of action. Hemisphere, Washington DC, pp. 195–213. https://doi.org/10.4324/9781315800820.
Jex, S.M., Bliese, P.D., Buzzell, S., Primeau, J., 2001. The impact of self-efficacy on stressor–strain relations: Coping style as an explanatory mechanism. J. Appl. Psychol. 86, 401–409. https://doi.org/10.1037/0021-9010.86.3.401.
Jex, S.M., Bliese, P.D., 1999. Efficacy beliefs as a moderator of the impact of work-related stressors: a multilevel study. J. Appl. Psychol. 84 (3), 349–361. https://doi.org/10.1037/0021-9010.84.3.349.
Johnston, A.C., Warkentin, M., 2010. Fear appeals and information security behaviors: an empirical study. MIS Q. 549–566. https://doi.org/10.2307/25750691.
Jones, G.R., 1986. Socialization tactics, self-efficacy, and newcomers' adjustments to organizations. Acad. Manag. J. 29 (2), 262–279. https://doi.org/10.5465/256188.
Joseph, D., Ang, S., 2001. The threat-rigidity model of professional obsolescence and its impact on occupational mobility behaviors of IT professionals. ICIS 2001 Proc. 73. ⟨http://aisel.aisnet.org/icis2001/73⟩.
Ketchen Jr, D.J., Palmer, T.B., 1999. Strategic responses to poor organizational performance: a test of competing perspectives. J. Manag. 25 (5), 683–706. https://doi.org/10.1177/01492063990250050.
Kinicki, A.J., Latack, J.C., 1990. Explication of the construct of coping with involuntary job loss. J. Vocat. Behav. 36 (3), 339–360. https://doi.org/10.1016/0001-8791(90)90036-2.
Knight, W., 2024. Amazon dreams of AI agents that do the shopping for you. WIRED. ⟨https://www.wired.com/story/amazon-ai-agents-shopping-guides-rufus⟩.
Korinek, A., Stiglitz, J.E., 2018. Artificial intelligence and its implications for income distribution and unemployment. In *The economics of artificial intelligence: An agenda*. University of Chicago Press, pp. 349–390.
Kraimer, M.L., Wayne, S.J., Liden, R.C., Sparrowe, R.T., 2005. The role of job security in understanding the relationship between employees' perceptions of temporary workers and employees' performance. J. Appl. Psychol. 90, 389–398. https://doi.org/10.1037/0021-9010.90.2.389.
Lazarus, R.S., 1991. Progress on a cognitive-motivational-relational theory of emotion. Am. Psychol. 46, 819–834. https://doi.org/10.1037/0003-066x.46.8.819.
Lazarus, R.S., Folkman, S., 1984. Stress, appraisal, and coping. Springer Publishing Company, New York.
Lazarus, R.S., Folkman, S., 1987. Transactional theory and research on emotions and coping. Eur. J. Personal. 1, 141–169. https://doi.org/10.1002/per.2410010304.
Lee, J., Peccei, R., 2007. Perceived organizational support and affective commitment: the mediating role of organization-based self-esteem in the context of job insecurity. J. Organ. Behav. 28 (6), 661–685. https://doi.org/10.1002/job.431.
Leung, E., Paolacci, G., Puntoni, S., 2018. Man versus machine: resisting automation in identity-based consumer behavior. J. Mark. Res. 55, 818–831. https://doi.org/10.1177/0022243718818423.
Liang, H., Xue, Y., 2009. Avoidance of information technology threats: a theoretical perspective. MIS Q. 1, 71–90. https://doi.org/10.2307/20650279.
Liang, H., Xue, Y., Pinsonneault, A., Wu, Y., 2019. What users do besides problem-focused coping when facing it security threats: an emotion-focused coping perspective. MIS Q. 43 (2), 373–394. https://doi.org/10.25300/misq/2019/14360.
Lichtenthaler, U., 2019. Extremes of acceptance: employee attitudes toward artificial intelligence. J. Bus. Strategy. https://doi.org/10.1108/jbs-12-2018-0204.
Lim, J., 2021. How AI algorithms are changing trading forever. Entrepreneur. ⟨https://www.entrepreneur.com/article/377985⟩.
Lim, S., Tai, K., 2014. Family incivility and job performance: a moderated mediation model of psychological distress and core self-evaluation. J. Appl. Psychol. 99, 351–359. https://doi.org/10.1177/0022243718818423.
Madani, M., Behzadi, M.M., Nabavi, S., 2022. The role of deep learning in advancing breast cancer detection using different imaging modalities: a systematic review. Cancers 14 (21), 5334. https://doi.org/10.3390/cancers14215334.

Makara-Studzińska, M., Golonka, K., Izydorczyk, B., 2019. Self-efficacy as a moderator between stress and professional burnout in firefighters. Int. J. Environ. Res. Public Health 16 (2), 183. https://doi.org/10.3390/ijerph16020183.

Makarius, E.E., Mukherjee, D., Fox, J.D., Fox, A.K., 2020. Rising with the machines:a sociotechnical framework for bringing artificial intelligence into the organization. J. Bus. Res. 120, 262–273. https://doi.org/10.1016/j.jbusres.2020.07.045.

Manyika, V.J., Chui, M., Miremadi, M., Bughin, J., George, K., Willmott, P., Dewhurst, M., 2017. A Future That Works: Automation, Employment and Productivity. McKinsey Global Institute Research. 〈https://www.jbs.cam.ac.uk/wp-content/uploads/2020/08/170622-slides-manyika.pdf〉.

Mathieu, J.E., Martineau, J.W., 2014. Individual and situational influences in training motivation. In *Improving training effectiveness in work organizations*. Psychology Press, pp. 205–234.

Maurer, T.J., 2001. Career-relevant learning and development, worker age, and beliefs about self-efficacy for development. J. Manag. 27 (2), 123–140. https://doi.org/10.1177/014920630102700201.

McGregor, H.A., Elliot, A.J., 2002. Achievement goals as predictors of achievement-relevant processes prior to task engagement. J. Educ. Psychol. 94 (2), 381. https://doi.org/10.1037//0022-0663.94.2.381.

Morgeson, F.P., Delaney-Klinger, K., Hemingway, M.A., 2005. The importance of job autonomy, cognitive ability, and job-related skill for predicting role breadth and job performance. J. Appl. Psychol. 90 (2), 399–406. https://doi.org/10.1037/0021-9010.90.2.399.

Morse, G., 2020. Harnessing Artificial Intelligence. Harv. Bus. Rev. 158–159.

Norman, P., Collins, S., Conner, M., Martin, R., Rance, J., 1995. Attributions, cognitions, and coping styles: teleworkers' reactions to work-related problems 1. J. Appl. Soc. Psychol. 25, 117–128. https://doi.org/10.1111/j.1559-1816.1995.tb01587.x.

Oldham, G.R., Kulik, C.T., Ambrose, M.L., Stepina, L.P., Brand, J.F., 1986. Relations between job facet comparisons and employee reactions. Organ. Behav. Hum. Decis. Process. 38 (1), 28–47. https://doi.org/10.1016/0749-5978(86)90024-5.

Pan, Y., Froese, F., Liu, N., Hu, Y., Ye, M., 2023. The adoption of artificial intelligence in employee recruitment: the influence of contextual factors. Int. J. Hum. Resour. Manag. 33 (6), 1125–1147. https://doi.org/10.1080/09585192.2021.1879206.

Parker, S.K., 1998. Enhancing role breadth self-efficacy: the roles of job enrichment and other organizational interventions. J. Appl. Psychol. 83 (6), 835–852. https://doi.org/10.1037/0021-9010.83.6.835.

Preacher, K.J., Zyphur, M.J., Zhang, Z., 2010. A general multilevel SEM framework for assessing multilevel mediation. Psychol. Methods 15 (3), 209–233. 〈https://doi.org/10.1037/a0020141〉.

Raisch, S., Krakowski, S., 2021. Artificial intelligence and management: the automation–augmentation paradox. Acad. Manag. Rev. 46 (1), 192–210. https://doi.org/10.5465/amr.2018.0072.

Ransbotham, S., Kiron, D., Khodabandeh, S., Fehling, R., LaFountain, B., Kiron, D., 2019. Winning with AI. MIT Sloan Rev. 〈https://sloanreview.mit.edu/projects/winning-with-ai/〉.

Reh, S., Tröster, C., Van Quaquebeke, N., 2018. Keeping future rivals down: temporal social comparison predicts coworker social undermining via future status threat and envy. J. Appl. Psychol. 103 (4), 399. https://doi.org/10.1037/apl0000281.

Roseman, I.J., Spindel, M.S., Jose, P.E., 1990. Appraisals of emotion-eliciting events: testing a theory of discrete emotions. J. Personal. Soc. Psychol. 59 (5), 899–915. https://doi.org/10.1037/0022-3514.59.5.899.

Rosenblatt, Z., Ruvio, A., 1996. A test of a multidimensional model of job insecurity: the case of Israeli teachers. J. Organ. Behav. 17 (S1), 587–605. https://doi.org/10.1002/(SICI)1099-1379(199612).

Schwarzer, R., Boehmer, S., Luszczynska, A., Mohamed, N.E., Knoll, N., 2005. Dispositional self-efficacy as a personal resource factor in coping after surgery. Personal. Individ. Differ. 39 (4), 807–818. https://doi.org/10.1016/j.paid.2004.12.016.

Shuck, B., Ghosh, R., Zigarmi, D., Nimon, K., 2013. The jingle jangle of employee engagement: further exploration of the emerging construct and implications for workplace learning and performance. Hum. Resour. Dev. Rev. 12 (1), 11–35. https://doi.org/10.1177/15344843124639.

Siemens, 2024. Generative artificial intelligence takes Siemens' predictive maintenance solution to the next level. Siemens Press Release. 〈https://press.siemens.com/global/en/pressrelease/generative-artificial-intelligence-takes-siemens-predictive-maintenance-solution-next〉.

Sinsky, C., Colligan, L., Li, L., Prgomet, M., Reynolds, S., Goeders, L., Blike, G., 2016. Allocation of physician time in ambulatory practice: a time and motion study in 4 specialties. Ann. Intern. Med. 165 (11), 753–760. https://doi.org/10.7326/M16-0961.

Skavronskaya, L., Moyle, B., Scott, N., 2020. The experience of novelty and the novelty of experience. Front. Psychol. 11, 322. https://doi.org/10.3389/fpsyg.2020.00322.

Smith, C.A., Ellsworth, P.C., 1985. Patterns of cognitive appraisal in emotion. J. Personal. Soc. Psychol. 48 (4), 813–838. https://doi.org/10.1037/0022-3514.48.4.813.

Staw, B.M., Sandelands, L.E., Dutton, J.E., 1981. Threat rigidity effects in organizational behavior: a multilevel analysis. Adm. Sci. Q. 26, 501–524. https://doi.org/10.2307/2392337.

Sterling, A., Boxall, P., 2013. Lean production, employee learning and workplace outcomes: a case analysis through the ability-motivation-opportunity framework. Hum. Resour. Manag. J. 23 (3), 227–240. https://doi.org/10.1111/1748-8583.12010.

Strich, F., Mayer, A.S., Fiedler, M., 2021. What do i do in a world of artificial intelligence? investigating the impact of substitutive decision-making ai systems on employees' professional role identity. J. Assoc. Inf. Syst. 22 (2), 304–324. https://doi.org/10.17705/1jais.00663.

Tambe, P., Cappelli, P., Yakubovich, V., 2019. Artificial intelligence in human resources management: challenges and a path forward. Calif. Manag. Rev. 61 (4), 15–42. https://doi.org/10.1177/0008125619867910.

Tong, S., Jia, N., Luo, X., Fang, Z., 2021. The Janus face of artificial intelligence feedback: deployment versus disclosure effects on employee performance. Strateg. Manag. J. 42 (9), 1600–1631. https://doi.org/10.2139/ssrn.3873901.

Tredinnick, L., 2017. Artificial intelligence and professional roles. Bus. Inf. Rev. 34, 37–41. https://doi.org/10.1177/0266382117692621.

Vander Elst, T., Van den Broeck, A., De Cuyper, N., De Witte, H., 2014. On the reciprocal relationship between job insecurity and employee well-being: mediation by perceived control? J. Occup. Organ. Psychol. 87 (4), 671–693. https://doi.org/10.1111/joop.12068.

VandeWalle, D., Brown, S.P., Cron, W.L., Slocum Jr, J.W., 1999. The influence of goal orientation and self-regulation tactics on sales performance: a longitudinal field test. J. Appl. Psychol. 84 (2), 249–259. https://doi.org/10.1037/0021-9010.84.2.249.

Von Hippel, W., Von Hippel, C., Conway, L., Preacher, K.J., Schooler, J.W., Radvansky, G.A., 2005. Coping with stereotype threat: denial as an impression management strategy. J. Personal. Soc. Psychol. 89 (1), 22–35. https://doi.org/10.1037/0022-3514.89.1.22.

Vroom, V., 1964. Work and Motivation. Wiley, New York.

Wejnert, B., 2002. Integrating models of diffusion of innovations: A conceptual framework. Annu. Rev. Sociol. 28, 297–326. https://doi.org/10.1146/annurev.soc.28.110601.141051.

Wilson, H.J., Daugherty, P.R., 2018. Collaborative intelligence: humans and AI are joining forces. Harv. Bus. Rev. 96, 114–123. 〈https://hometownhealthonline.com/wp-content/uploads/2019/02/ai2-R1804J-PDF-ENG.pdf〉.

Zhang, F., Parker, S.K., 2019. Reorienting job crafting research: a hierarchical structure of job crafting concepts and integrative review. J. Organ. Behav. 40 (2), 126–146. https://doi.org/10.1002/job.2332.

Zhou, J., Shin, S.J., Cannella Jr, A.A., 2008. Duan self-perceived creativity after mergers and acquisitions: interactive effects of threat—opportunity perception, access to resources, and support for creativity. J. Appl. Behav. Sci. 44, 397–421. https://doi.org/10.1177/0021886308328010.

Zuboff, S., 1988. In the age of the smart machine: The future of work and power. Basic Books, Inc, New York.